Big science and peer review, a game theoretical model

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1 Introduction

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The publication game involves two classes of players: a single journal (J) and a finite population of authors $\mathcal{A} = \{1, 2, \ldots, N\}$. We treat N as a finite exogenous parameter. This finite-population approach allows us to capture the discrete congestion dynamics and "tragedy of the commons" effects inherent in resource-constrained peer review systems, as analyzed in the economic literature on academic publishing (Cotton, 2013; Azar, 2005). Throughout, we use V and M to denote the realised *counts* of submitted and peer-accepted papers, respectively.

Quality generation. In each round, each author generates a manuscript with binary intrinsic quality $q \in \{0, 1\}$, where $q = 1$ denotes a "good" paper and $q = 0$ a "bad" paper. Quality is generated i.i.d. across authors with

$$\Pr(q = 1) = \alpha, \qquad \Pr(q = 0) = 1 - \alpha, \tag{1}$$

where $\alpha \in (0, 1)$ is an exogenous productivity parameter reflecting the baseline scientific capability of the community (Ellison, 2002).

Author strategies (S_A). Each author chooses a submission rule $s_A \in S_A$ that conditions on realised quality. Consistent with signaling models in scientific communication (Tiokhin et al., 2021), we consider four pure strategies:

- **Only Good (OG):** submit iff $q = 1$;
- **Always Submit (AS):** submit regardless of q (strategic spamming);
- **Only Bad (OB):** submit iff $q = 0$ (included for completeness);
- **Never Submit (NS):** never submit.

Equivalently, s_A can be represented by (p_g, p_b), the submission probabilities for good and bad papers:

$$(p_g, p_b) = \begin{cases} (1, 0) & \text{if } s_A = \text{OG} \\ (1, 1) & \text{if } s_A = \text{AS} \\ (0, 1) & \text{if } s_A = \text{OB} \\ (0, 0) & \text{if } s_A = \text{NS}. \end{cases} \tag{2}$$

Journal strategies (S_J). The journal commits to an editorial policy $s_J \in S_J$ that maps the perceived quality signal $\hat{q} \in \{0, 1\}$ (generated by noisy peer review, see Section **??**) into an accept/reject decision:

- **Only Good (OG):** accept iff $\hat{q} = 1$;
- **Always Accept (AA):** accept all submissions (ignores $\hat{q}$);
- **Only Bad (OB):** accept iff $\hat{q} = 0$;
- **Reject All (AR):** reject all submissions (ignores $\hat{q}$).

Exogenous parameters and endogenous variables. To clarify the mechanics of congestion, we distinguish between fixed environmental constraints and emergent outcomes:

- **Exogenous parameters:** Population size (N), productivity (α), review capacity (K), review noise parameters (ϵ, λ), and payoff primitives (c, r, B, D).
- **Endogenous variables:** Submission volume (V), peer-accepted volume (M), the rationing factor (ρ), and the population state x (strategy shares) governing the equilibrium distribution.

We consider a publication game involving a finite population of N authors and a single academic journal. Our framework aligns with established game-theoretic models of scientific publishing Thurner and Hanel (2011), while specifically building on the incentive structures analyzed by Zollman et al. Zollman et al. (2024). Unlike previous static models, we extend the analysis to encompass system-wide congestion. The fundamental unit of the game is a research paper, which possesses a binary intrinsic quality $q \in \{0, 1\}$. A paper is defined as "good" ($q = 1$) or "bad" ($q = 0$). In each period, authors make submission decisions, and the journal enforces quality control standards subject to a physical capacity constraint.

The publication process unfolds in a strictly sequential timeline, as illustrated in Figure 1:

1. **Submission (Generation of V):** Based on their strategies, authors decide whether to submit their work. The aggregate collection of all submitted manuscripts forms the *Submission Pool*, denoted by $V \leq N$.
2. **Peer Review (Generation of M):** The journal applies a peer review process to filter the submission pool. Acknowledging the inherent unreliability of assessment documented in empirical studies Bornmann (2011), we model reviewers as imperfect classifiers. They are subject to Type I errors (False Positives, rate λ) and Type II errors (False Negatives, rate ϵ). Papers that successfully pass this screening form the set of *Peer-accepted papers*, denoted by M. It is important to note that membership in M signifies that a paper is "scientifically acceptable," but not necessarily "published."
3. **Capacity Check and Storage (K):** The journal has a fixed resource limit or *Journal Storage*, denoted by K. This constraint reflects the physical limits on publication slots or the administrative burden of processing manuscripts Kovanis et al. (2016).
4. **Final Acceptance (ρ):** The final publication decision depends on the relationship between the volume of peer-accepted papers (M) and the journal storage (K). We define ρ as the *probability of being accepted conditional on being peer-accepted*. This variable captures the congestion effect:

$$\rho = \begin{cases} 1 & \text{if } M \leq K \quad \text{(No Congestion)} \\ K/M & \text{if } M > K \quad \text{(Rationing triggered)} \end{cases} \tag{3}$$

In this framework, a paper is published if and only if it passes two hurdles: the quality hurdle (peer review) and the congestion hurdle (storage constraint).

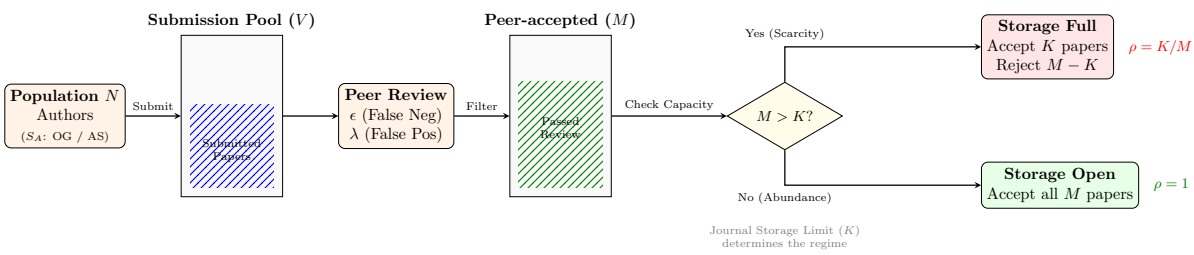


Figure 1: **Schematic representation of the Threshold Competition Model.** The process unfolds in stages: (1) Authors from population N submit papers into pool V; (2) Peer review filters submissions into a set of peer-accepted papers M; (3) The volume M is compared against the journal storage capacity K. If $M > K$, a rationing mechanism is triggered, reducing the conditional acceptance probability ρ below 1.

2.3 Review noise and pass-review probabilities

Peer review acts as a stochastic filter between the submission volume V and the peer-accepted volume M. Throughout, *pass* refers to passing the review filter and entering the peer-accepted pool M (not final publication, which is governed by the rationing factor ρ in Section 2.2).

Noise parameters (ϵ, λ). Consistent with empirical meta-analyses on review reliability (Bornmann et al., 2010; Cicchetti, 1991), we assume reviewers are imperfect classifiers. Let $\hat{q} \in \{0, 1\}$ denote the perceived quality signal for a paper with true quality $q \in \{0, 1\}$. We define:

$$\epsilon = \Pr(\hat{q} = 0 \mid q = 1), \qquad \lambda = \Pr(\hat{q} = 1 \mid q = 0), \tag{4}$$

where ϵ is the false negative rate (acting as a tax on honesty) and λ the false positive rate (acting as an incentive for opportunism), a structure supported by empirical studies on scientific gatekeeping (Siler et al., 2015).

Conditional pass probabilities. Given a journal policy s_J, define the probability of acceptance conditional on true quality as:

$$\pi_q(s_J) \equiv \Pr(\text{pass} \mid q, s_J), \qquad q \in \{0, 1\}. \tag{5}$$

Under the journal strategies defined in Section 2.2, and aligned with theoretical models of gatekeeper selection (Muller-Itten, 2022), the acceptance vector (π_1, π_0) is given by:

$$(\pi_1(s_J), \pi_0(s_J)) = \begin{cases} (1-\epsilon,\ \lambda) & \text{if } s_J = \text{OG} \\ (1,\ 1) & \text{if } s_J = \text{AA} \\ (\epsilon,\ 1-\lambda) & \text{if } s_J = \text{OB} \\ (0,\ 0) & \text{if } s_J = \text{AR}. \end{cases} \tag{6}$$

Per-author pass-review probability $(\mathcal{P})$. Let $(p_g(s_A), p_b(s_A))$ denote the submission probabilities conditional on realised quality. We define $\mathcal{P}(s_A, s_J)$ as the *ex-ante* probability that an author playing s_A contributes a paper to the peer-accepted pool M under journal policy s_J:

$$\mathcal{P}(s_A, s_J) = \alpha\, p_g(s_A)\, \pi_1(s_J) + (1-\alpha)\, p_b(s_A)\, \pi_0(s_J). \tag{7}$$

Below we enumerate the effective passing probabilities for all author strategies under each journal regime.

Case 1: Standard Review (Journal plays Only Good). This is the baseline scenario where $\pi_1 = 1 - \epsilon$ and $\pi_0 = \lambda$.

$$\mathcal{P}(\text{OG}, \text{OG}) = \alpha(1-\epsilon), \tag{8a}$$
$$\mathcal{P}(\text{AS}, \text{OG}) = \alpha(1-\epsilon) + (1-\alpha)\lambda, \tag{8b}$$
$$\mathcal{P}(\text{OB}, \text{OG}) = (1-\alpha)\lambda, \tag{8c}$$
$$\mathcal{P}(\text{NS}, \text{OG}) = 0. \tag{8d}$$

Case 2: Permissive (Journal plays Always Accept). The screening is removed $(\pi_1 = 1, \pi_0 = 1)$, making acceptance depend solely on submission.

$$\mathcal{P}(\text{OG}, \text{AA}) = \alpha, \tag{9a}$$
$$\mathcal{P}(\text{AS}, \text{AA}) = \alpha + (1-\alpha) = 1, \tag{9b}$$
$$\mathcal{P}(\text{OB}, \text{AA}) = 1 - \alpha, \tag{9c}$$
$$\mathcal{P}(\text{NS}, \text{AA}) = 0. \tag{9d}$$

Case 3: Inverse Selection (Journal plays Only Bad). The journal accepts papers perceived as bad $(\hat{q} = 0)$. Good papers pass only via false negatives $(\pi_1 = \epsilon)$, and bad papers pass via true negatives $(\pi_0 = 1 - \lambda)$.

$$\mathcal{P}(\text{OG}, \text{OB}) = \alpha\epsilon, \tag{10a}$$
$$\mathcal{P}(\text{AS}, \text{OB}) = \alpha\epsilon + (1-\alpha)(1-\lambda), \tag{10b}$$
$$\mathcal{P}(\text{OB}, \text{OB}) = (1-\alpha)(1-\lambda), \tag{10c}$$
$$\mathcal{P}(\text{NS}, \text{OB}) = 0. \tag{10d}$$

Case 4: Closed (Journal plays Reject All). No papers enter the peer-accepted pool ($\pi_1 = 0, \pi_0 = 0$).

$$\mathcal{P}(s_A, \mathrm{AR}) = 0 \quad \forall s_A \in S_A. \tag{11}$$

Aggregate peer-accepted volume ($E[M]$). To determine the system-wide congestion, we aggregate these individual passing probabilities across the entire population, a method consistent with burden estimation in peer review (Kovanis et al., 2016). Let $\mathbf{x} = \{x_{s_A} \mid s_A \in S_A\}$ denote the **population state**, where $x_{s_A} \in [0, 1]$ represents the proportion of authors adopting strategy s_A, satisfying $\sum_{s_A \in S_A} x_{s_A} = 1$.

The expected total volume of peer-accepted papers, $E[M]$, is the sum of contributions from all strategy groups:

$$E[M] = \sum_{i=1}^{N} \mathcal{P}(s_{A,i}, s_J) = N \sum_{s_A \in S_A} x_{s_A} \, \mathcal{P}(s_A, s_J). \tag{12}$$

This aggregate volume $E[M]$ is the critical endogenous variable that interacts with the journal capacity K to determine the rationing probability ρ defined in Eq. (1).

2.4 Capacity constraint and rationing probability (ρ)

A key feature of the model is the separation between *scientific merit* (passing peer review and entering the set of peer-accepted papers) and *publication slots* (capacity-constrained publication).

Journal capacity (K). Let $K \in \mathbb{N}^+$ denote the journal's exogenous publication capacity: at most K papers can be published per cycle due to limited pages and editorial resources.

Uniform rationing and the rationing probability (ρ). Let M be the realised number of peer-accepted papers in a given round. If $M \leq K$, all peer-accepted papers are published. If $M > K$, the journal applies *uniform rationing*: it selects exactly K papers uniformly at random from the M peer-accepted papers. Hence, the conditional publication probability for a peer-accepted paper is determined by the realised congestion:

$$\rho(M) \equiv \Pr(\text{published} \mid \text{pass}) = \min\left(1, \frac{K}{M}\right) = \begin{cases} 1 & \text{if } M \leq K, \\ \frac{K}{M} & \text{if } M > K. \end{cases} \tag{13}$$

Ex-post vs. ex-ante congestion. Across rounds, the peer-accepted count M is stochastic because it aggregates individual submission and review outcomes. Authors therefore form expectations over congestion given the population state $\mathbf{x}$ and journal policy s_J, implying an expected rationing factor:

$$\mathbb{E}[\rho] = \mathbb{E}\left[\min\left(1, \frac{K}{M}\right)\right]. \tag{14}$$

Crucially, increases in low-quality or opportunistic submissions raise the expected volume $\mathbb{E}[M]$ and mechanically reduce $\mathbb{E}[\rho]$, generating a negative congestion externality on all authors.

The journal chooses one of four editorial strategies:

$$S_J = \{OA, OG, OB, NS\},$$

defined as acceptance rules based on perceived paper quality:

- OA (*Open/Always Accept*): accept all submissions;
- OG (*Only Good*): accept submissions perceived as good;
- OB (*Only Bad*): accept submissions perceived as bad;
- NS (*No Select/Reject All*): reject all submissions.

2.5 Payoff analysis

In this subsection, we derive the expected payoffs for both authors and the journal. The introduction of the capacity constraint K couples the payoffs: an individual's utility depends not only on their own strategy but also on the aggregate congestion generated by the population.

2.5.1 Notation and general framework

Table 1: Summary of key parameters and variables used in the payoff analysis.

Symbol	Description
Exogenous Parameters (Fixed)	
N	Total number of authors
K	Journal capacity (maximum publication slots per cycle)
α	Probability that an author produces a good paper
ϵ	False negative rate (Prob. of rejecting a good paper)
λ	False positive rate (Prob. of accepting a bad paper)
r	Reward for a published paper (Author)
c	Cost of submission (Author)
B	Benefit for publishing a good paper (Journal)
D	Penalty for publishing a bad paper (Journal)
C_J	Unit cost of peer review (per submission)
Endogenous Variables (Dependent on Strategy)	
V	Total number of submissions (Submission Pool)
M	Number of papers passing peer review (Perception Pool)
Y	Number of papers finally published ($Y \leq K$)
$\bar{\rho}$	Expected rationing probability ($\mathbb{E}[\min(1, K/M)]$)
x	Proportion of authors adopting a specific strategy (Population State)

We first establish a unified notation. To ensure analytical tractability while capturing the congestion effect, we employ a mean-field approximation where players base their decisions on the **expected** state of the system.

Submission and publication probabilities. For an author playing strategy $s_A \in S_A$, we define the unconditional probability of submitting a paper as $P_{\text{sub}}(s_A)$. Based on the submission vectors (p_g, p_b) defined in Section 2.2, this is:

$$P_{\text{sub}}(s_A) = \alpha p_g(s_A) + (1-\alpha) p_b(s_A). \tag{15}$$

Recall from Section 2.3 that $\mathcal{P}(s_A, s_J)$ denotes the probability of *entering the peer-accepted pool* M. However, entering M does not guarantee publication. Let $\bar{\rho}$ denote the **expected rationing probability**:

$$\bar{\rho} \equiv \mathbb{E}[\rho] = \mathbb{E}\left[\min\left(1, \frac{K}{M}\right)\right]. \tag{16}$$

The joint probability of a paper being finally published is the product of passing review and passing the capacity check: $\bar{\rho} \cdot \mathcal{P}(s_A, s_J)$.

Volumes: Submitted (V), Passed (M), and Published (Y). It is crucial to distinguish between papers that pass peer review (M) and those that are finally published (Y). Let V_G and V_B be the expected volumes of good and bad submissions generated by the population. Let M_G and M_B be the expected volumes of papers entering the peer-accepted pool. We define Y_G and Y_B as the expected volumes of **finally published** papers:

$$Y_G = \bar{\rho} M_G, \quad Y_B = \bar{\rho} M_B. \tag{17}$$

2.5.2 General payoff functions

Author's utility (U_A). Let r be the reward for a published paper and c be the cost of submission. The expected payoff for an author playing s_A against a journal s_J is the expected reward from publication minus the reliable cost of submission:

$$U_A(s_A, s_J) = r \cdot \underbrace{\bar{\rho} \cdot \mathcal{P}(s_A, s_J)}_{\text{Prob(Published)}} - c \cdot \underbrace{P_{\text{sub}}(s_A)}_{\text{Prob(Submitted)}} \tag{18}$$

Journal's utility (U_J). Let B be the benefit of publishing a good paper, D be the penalty for a bad paper, and C_J be the unit cost of reviewing a submission. The journal's utility depends on the *published* volumes (Y_G, Y_B) and the *submitted* volume $(V = V_G + V_B)$:

$$U_J(s_J) = B \cdot Y_G - D \cdot Y_B - C_J \cdot V \tag{19}$$

Substituting $Y = \bar{\rho}M$, we obtain the working form of the journal's payoff:

$$U_J(s_J) = \bar{\rho}\,[B \cdot M_G(s_J) - D \cdot M_B(s_J)] - C_J \cdot V \tag{20}$$

Remark: Eq. (20) assumes $\mathbb{E}[\rho M] \approx \mathbb{E}[\rho]\mathbb{E}[M]$. This mean-field approximation is standard in evolutionary game theory for large finite populations.

2.5.3 Derivation of strategy-specific payoffs

We now explicitly list the payoff functions for the 8 primary scenarios (4 Author strategies and 4 Journal strategies).

Author payoffs (under Journal Strategy s_J = OG). In the standard academic context, the journal plays **Only Good (OG)**. The review filter applies $(\pi_1, \pi_0) = (1 - \epsilon, \lambda)$. Substituting the specific $\mathcal{P}$ and P_{sub} values into Eq. (18), we derive:

$$U_A(\text{OG}, \text{OG}) = \alpha\,[r\bar{\rho}(1 - \epsilon) - c] \tag{21}$$
$$U_A(\text{AS}, \text{OG}) = r\bar{\rho}\,[\alpha(1 - \epsilon) + (1 - \alpha)\lambda] - c \tag{22}$$
$$U_A(\text{OB}, \text{OG}) = (1 - \alpha)\,[r\bar{\rho}\lambda - c] \tag{23}$$
$$U_A(\text{NS}, \text{OG}) = 0 \tag{24}$$

Journal payoffs (under Population State x). The journal chooses a strategy s_J to maximize utility given the incoming flow of submissions. Let $V_G(\mathbf{x})$ and $V_B(\mathbf{x})$ be the volumes of good and bad submissions determined by the population state.

1. Only Good (OG): The journal filters papers. The passed volumes are $M_G = V_G(1 - \epsilon)$ and $M_B = V_B\lambda$.

$$U_J(\text{OG}) = \bar{\rho}\,[BV_G(1 - \epsilon) - DV_B\lambda] - C_J(V_G + V_B) \tag{25}$$

2. Always Accept (AA): The journal accepts everything $(M_G = V_G, M_B = V_B)$. Congestion is maximized here, so $\bar{\rho}$ will be minimal.

$$U_J(\text{AA}) = \bar{\rho}\,[BV_G - DV_B] - C_J(V_G + V_B) \tag{26}$$

3. Only Bad (OB): The journal applies inverse selection $(M_G = V_G\epsilon, M_B = V_B(1 - \lambda))$.

$$U_J(\text{OB}) = \bar{\rho}\,[BV_G\epsilon - DV_B(1 - \lambda)] - C_J(V_G + V_B) \tag{27}$$

4. Reject All (AR): The journal publishes nothing $(Y = 0)$. Review costs are still incurred for processed submissions.

$$U_J(\text{AR}) = -C_J(V_G + V_B) \quad \text{(normalized to 0 in dominance analysis)} \tag{28}$$

2.5.4 Summary of transition probabilities

For completeness, Table 2 summarizes the effective probability $\mathcal{P}(s_A, s_J)$ of entering the peer-accepted pool for any strategy pair. These values are the key inputs for the payoff equations.

Table 2: Matrix of Effective Passing Probabilities $\mathcal{P}(s_A, s_J)$. This represents the probability that an author's paper enters the peer-accepted pool M (before rationing).

Author \ Journal	**Only Good (OG)**	**Always Accept (AA)**	**Only Bad (OB)**	**Reject All (AR)**
OG (α good)	$\alpha(1-\epsilon)$	α	$\alpha\epsilon$	0
AS (All)	$\alpha(1-\epsilon)+(1-\alpha)\lambda$	1	$\alpha\epsilon+(1-\alpha)(1-\lambda)$	0
OB ($1-\alpha$ bad)	$(1-\alpha)\lambda$	$1-\alpha$	$(1-\alpha)(1-\lambda)$	0
NS (None)	0	0	0	0

2.6 Dominance analysis and reduced game

Before analyzing the evolutionary dynamics, we simplify the strategy space by iteratively eliminating strictly dominated strategies. We conduct this analysis under the standard **mean-field approximation**, holding the aggregate congestion factor $\bar{\rho}$ fixed. The elimination proceeds in two stages: first, we identify strategies that are dominated purely by the structural logic of the game (independent of specific cost-benefit inequalities); second, we introduce parameter constraints to eliminate strategies based on economic trade-offs.

2.6.1 Stage 1: Structural elimination (Unconditional dominance)

We first establish dominance relations that hold under weak axiomatic assumptions regarding the nature of peer review and scientific value.

Elimination of Journal "Only Bad" (OB). The strategy **Only Bad (OB)** represents an inverse selection policy. We compare it to **Only Good (OG)**. Using the payoff functions derived in Section 2.5, the difference in expected utility is:

$$\begin{aligned}\Delta U_J &= U_J(\text{OG}) - U_J(\text{OB}) \\ &= \bar{\rho}\left[BV_G(1-\epsilon) - DV_B\lambda\right] - \bar{\rho}\left[BV_G\epsilon - DV_B(1-\lambda)\right] \\ &= \bar{\rho}\left[BV_G(1-2\epsilon) + DV_B(1-2\lambda)\right] \end{aligned} \tag{29}$$

Condition: We assume peer review is **informative**, defined as performing better than random guessing:

$$\epsilon < 0.5 \quad \text{and} \quad \lambda < 0.5. \tag{30}$$

Under this definition, both $(1-2\epsilon)$ and $(1-2\lambda)$ are strictly positive. Since $B, D, V_G, \bar{\rho} > 0$, it follows that $\Delta U_J > 0$. Thus, **Journal OB is strictly dominated by OG** solely by virtue of the definition of peer review quality, regardless of the magnitude of rewards or penalties.

Elimination of Author "Only Bad" (OB). Consider the choice between **Always Submit (AS)** and **Only Bad (OB)**. AS submits both paper types ($q \in \{0,1\}$), while OB submits only bad papers ($q = 0$). The payoff difference is the marginal utility of submitting a good paper:

$$\begin{aligned}\Delta U_A &= U_A(\text{AS}, \text{OG}) - U_A(\text{OB}, \text{OG}) \\ &= [U_A(q=1) + U_A(q=0)] - U_A(q=0) \\ &= U_A(q=1) = \alpha\left[r\bar{\rho}(1-\epsilon) - c\right] \end{aligned} \tag{31}$$

Condition: We assume the **Axiom of Value**: in a functional scientific system, a correctly identified high-quality paper yields a positive net return (i.e., $r(1-\epsilon) > c$). If this axiom holds, then $\Delta U_A > 0$ for any non-zero congestion $\bar{\rho}$. Thus, **Author OB is strictly dominated by AS** due to monotonicity: a rational author will never withhold a high-quality paper while simultaneously submitting a low-quality one.

2.6.2 Parameter constraints for a non-trivial game

Having eliminated the structurally irrational strategies, we now focus on the remaining set. To ensure the game yields interesting evolutionary dynamics (avoiding trivial collapse or utopian perfection), we define the "viable region" of the parameter space using the **abundance benchmark** ($\bar{\rho} = 1$).

1. **Viability Constraint (Honesty is profitable):**

$$r(1-\epsilon) > c \tag{32}$$

This formally restates the Axiom of Value. It ensures that without congestion, the honest strategy (OG) dominates silence (NS).

2. **Profitability Constraint (Spamming is profitable):**

$$\lambda r > c \tag{33}$$

This constraint drives the congestion dilemma. It implies that in a frictionless system ($\bar{\rho} = 1$), the expected reward of a bad paper exceeds the submission cost. If $c > \lambda r$, bad papers are never submitted, and the game trivially converges to perfect honesty.

3. **Reputation Constraint (Screening is preferred):**

$$DV_B(1-\lambda) > BV_G\epsilon \tag{34}$$

This condition ensures the journal prefers screening (OG) over blind acceptance (AA). It states that the aggregate penalty avoided by rejecting bad papers outweighs the benefit lost from accidentally rejecting good ones.

2.6.3 Stage 2: Conditional elimination (Economic dominance)

We now eliminate strategies based on the specific economic trade-offs defined above.

Dominance of Journal "Always Accept" (AA). We compare **Only Good (OG)** with **Always Accept (AA)**. Assuming the processing cost C_J applies to all submissions (Processing Cost Model), the difference depends on the net value of papers rejected by OG but accepted by AA:

$$\begin{aligned} U_J(\text{OG}) - U_J(\text{AA}) &= \bar{\rho}\left[(BM_G^{\text{OG}} - DM_B^{\text{OG}}) - (BM_G^{\text{AA}} - DM_B^{\text{AA}})\right] \\ &= \bar{\rho}\left[D(V_B - M_B^{\text{OG}}) - B(V_G - M_G^{\text{OG}})\right] \\ &= \bar{\rho}\left[DV_B(1-\lambda) - BV_G\epsilon\right] \end{aligned} \tag{35}$$

Under the **Reputation Constraint** (34), the penalty avoided (first term) exceeds the benefit lost (second term). Thus, **AA is strictly dominated by OG**.

Dominance of Journal "Reject All" (AR). Strategy AR yields a normalized payoff of 0 (or $-C_J V$). Since OG generates positive utility from good papers ($B > 0$) and filters bad ones, OG dominates AR provided the net value of the peer-accepted pool is positive: $BM_G > DM_B$. We assume the journal operates in a parameter region where valid science is net positive.

Dominance of Author "Never Submit" (NS). Comparing **Only Good (OG)** with **Never Submit (NS)** ($U_A = 0$):

$$U_A(\text{OG}, \text{OG}) - U_A(\text{NS}, \text{OG}) = \alpha\left[r\bar{\rho}(1-\epsilon) - c\right] \tag{36}$$

Under the **Viability Constraint** (32), this difference is positive as long as the congestion does not exceed the critical threshold $\bar{\rho} > \frac{c}{r(1-\epsilon)}$. In the viable regime, **NS is strictly dominated by OG** (and by AS).

2.6.4 The reduced evolutionary game

Following the two-stage elimination process, the complex multi-strategy interaction reduces to a concise 2×1 asymmetric game.

1. **The Journal** is a fixed institutional player locked into the **Only Good (OG)** strategy. It acts as the environment, applying a filter (ϵ, λ) and a capacity constraint K.

2. **The Author Population** evolves between two surviving pure strategies:

   - **Only Good (OG) [Cooperators]:** Authors who submit only high-quality work ($q = 1$).

- **Always Submit (AS) [Defectors]:** Authors who submit everything ($q \in \{0, 1\}$), effectively spamming the journal with bad papers to exploit λ.

We define the **population state** $x(t) \in [0, 1]$ as the fraction of authors adopting the **Only Good** strategy. The evolutionary dynamics of the system are thus entirely determined by the time evolution of x.

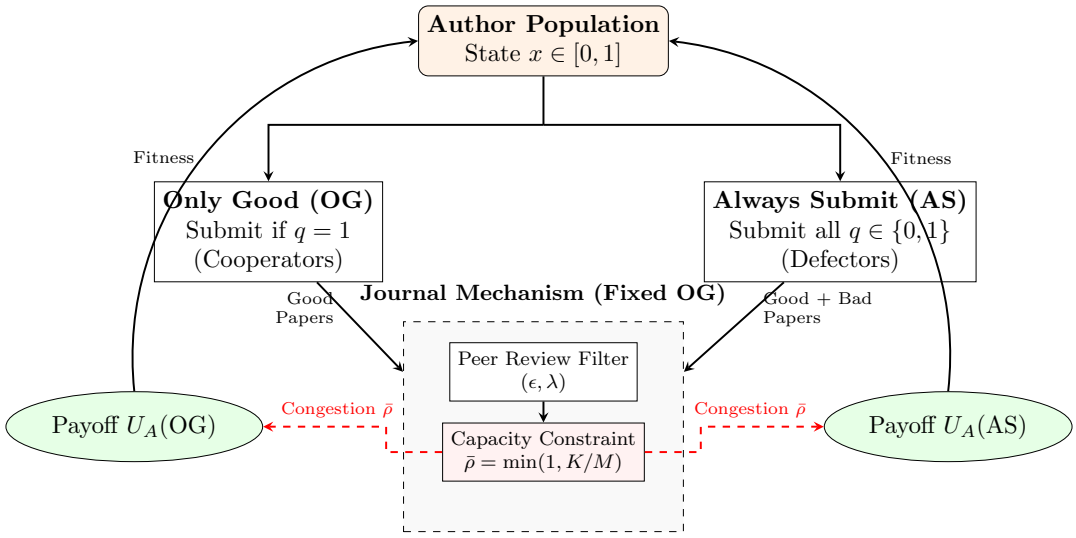


Figure 2: **Structure of the Reduced Evolutionary Game.** The author population evolves between two strategies: Only Good (OG) and Always Submit (AS). The Journal acts as a fixed mechanism consisting of a peer review filter and a capacity constraint. Submissions from AS authors increase the total volume M, triggering the capacity constraint. The resulting **Congestion Externality** ($\bar{\rho}$) reduces the publication probability for all papers, feeding back into the payoffs and driving the evolutionary dynamics.

3 Replicator Dynamics and Bifurcation Analysis

3.1 Reduction to a Two-Strategy System and Congestion Model

Following the elimination of strictly dominated strategies (specifically {Only Bad} and {Never Submit}), the game is reduced to a dynamic contest between two Author strategies, while the Journal plays a fixed strategy of *Only Good*. We define the population state $x \in [0, 1]$ as follows:

- x: The proportion of Authors playing **{Only Good}**.
- $1 - x$: The proportion of Authors playing **{Always Submit}**.

Since the Journal's strategy is fixed, the evolutionary dynamics are driven entirely by the payoff difference between the Author's two remaining strategies. To allow for the survival of the *Always Submit* strategy and the emergence of complex dynamics, we assume the **Profitability Condition** ($\lambda B > c$) holds. This implies that submitting a "Bad" paper is profitable *if* the journal fails to screen it.

3.1.1 Physical Definitions and Variables

We introduce a finite resource constraint on the Journal to model the **Congestion Effect**. The parameters are defined as follows:

- N: Total population of authors.
- K: Journal processing capacity (total review effort available).
- C_J: Cost per review. Thus, K/C_J represents the maximum number of papers the journal can effectively screen.
- α: The prior probability that an author holds a "Good" paper.

- B: Benefit (Reward) of acceptance.
- C: Cost of submission.
- D: Reputational damage penalty for a rejected "Bad" paper.
- λ: The discount factor for a "Bad" paper (Payoff $= \lambda B$).
- ϵ: A friction term to prevent singularities.

3.2 Derivation of the Polynomial Dynamics

To analyze the stability of the system, we express the rate of change $\frac{dx}{dt}$ as a function of the submission volume and screening effectiveness.

3.2.1 Volume and Screening Functions

The total submission volume $V(x)$ depends on the strategy distribution. "Only Good" authors submit with probability α, while "Always Submit" authors submit with probability 1:

$$V(x) = N[\alpha x + (1-x)] = N[1-(1-\alpha)x] \tag{37}$$

Note that $V(x)$ is linear and decreasing in x; as honesty increases, junk submissions decrease.

The screening probability $\sigma(x)$ is endogenous and depends on the ratio of capacity to volume:

$$\sigma(x) = \frac{K}{C_J \cdot V(x) + \epsilon} \tag{38}$$

This inverse relationship creates the nonlinearity required for bistability.

3.2.2 Utility Functions and Replicator Equation

The evolutionary dynamics depend on the expected utility of submitting a "Bad" paper, $E[U_{Bad}]$.

- If screened (probability σ): Payoff is $-C-D$.
- If unscreened (probability $1-\sigma$): Payoff is $\lambda B - C$.

The expected utility is:

$$\begin{aligned} E[U_{Bad}](x) &= \sigma(x)(-C-D) + (1-\sigma(x))(\lambda B - C) \\ &= \sigma(x)(-D-\lambda B) + (\lambda B - C) \end{aligned} \tag{39}$$

Substituting this into the standard replicator equation $\frac{dx}{dt} = -x(1-x)(1-\alpha)E[U_{Bad}]$, and clearing the denominator $(C_J V(x)+\epsilon)$, we derive the polynomial $P(x)$ that characterizes the sign of the dynamics:

$$P(x) = \underbrace{x(1-x)}_{\text{Deg 2}} \cdot \underbrace{\Big[K(-D-\lambda B) + (\lambda B - C)(C_J N(1-(1-\alpha)x) + \epsilon)\Big]}_{\text{Deg 1}} \tag{40}$$

3.2.3 Degree Analysis and Root Configuration

The polynomial $P(x)$ is of **Degree 3 (Cubic)**. This is confirmed by the product of the quadratic replicator term $x(1-x)$ and the linear volume term nested within the utility function. Consequently, the system admits up to three real roots in $[0,1]$:

1. $x = 0$: **Full Corruption** (Stable if congestion makes cheating profitable).
2. $x = 1$: **Ideal Honesty** (Stable if low volume makes screening strict).
3. $x = x^*$: An internal **Tipping Point** (Unstable).

The analytical form of the internal root is derived as:

$$x^* = \frac{1}{1-\alpha}\left[1 + \frac{\epsilon}{C_J N} - \frac{K(D+\lambda B)}{C_J N(\lambda B - C)}\right] \tag{41}$$

3.3 Numerical Simulation and Parameter Analysis

3.3.1 Case 1: The Failure of Congestion ($K = 500$)

In the initial simulation with $N = 1000$ and $K = 500$, the journal capacity was sufficient to screen all papers even at maximum volume (N). Mathematically, this resulted in $x^* < 0$. Under these conditions, the effective screening rate $\sigma \approx 1$ rendered cheating globally unprofitable ($U_{Bad} < 0$), collapsing the game to a single stable equilibrium at $x = 1$.

3.3.2 Case 2: The Emergence of Bistability ($K = 100$)

By reducing capacity to $K = 100$, we induced the congestion effect. At $x = 0$, high volume overwhelms the journal (σ is low), satisfying the profitability condition $\lambda B > C$. At $x = 1$, volume is low, σ is high, and cheating is punished. This trade-off creates the internal root x^*.

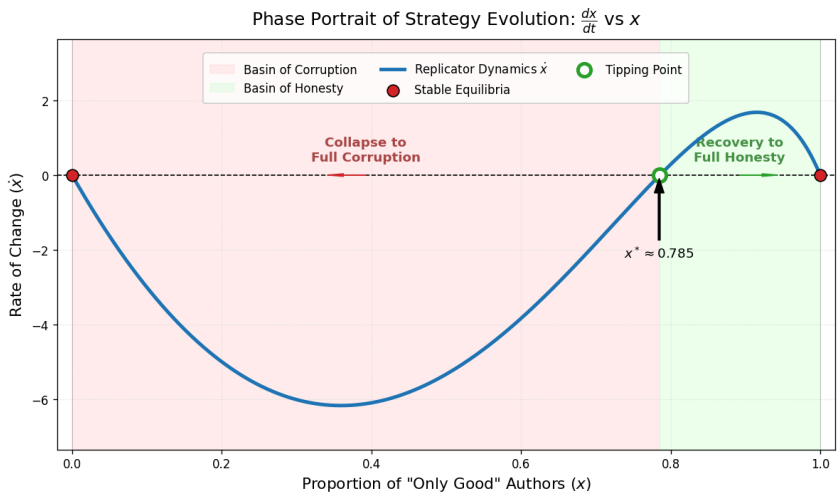


Figure 3: Replicator Dynamics $\frac{dx}{dt}$ vs x with $K = 100$. The system exhibits bistability. The internal root $x^* \approx 0.786$ acts as an unstable repeller. If the initial proportion of honest authors $x < x^*$, the system collapses to full corruption ($x = 0$). If $x > x^*$, it converges to full honesty ($x = 1$).

Figure 3 visualizes this dynamic. The existence of x^* in $(0, 1)$ confirms that the "Always Submit" strategy is viable only when the journal is congested.

3.3.3 Analytical Proof of Equilibrium Instability via Monotonicity

To rigorously characterize the stability of the internal equilibrium x^*, we analyze the first derivative of the evolutionary velocity field with respect to the population state x. The sign of the replicator dynamic equation $\dot{x}$ is determined strictly by the payoff differential between the "Only Good" and "Always Submit" strategies. Since the payoff for honest authors is strictly positive under the viability constraint, the system's trajectory is governed by the sign of the net disadvantage of the "Always Submit" strategy. We define the stability indicator function $\Psi(x)$ as the negative expected utility of submitting a "Bad" paper:

$$\Psi(x) = -E[U_{Bad}](x) = \sigma(x)(D + \lambda B) - (\lambda B - C) \tag{42}$$

The stability of any fixed point x^* where $\Psi(x^*) = 0$ depends on the slope $\Psi'(x^*)$. A positive slope implies that the dynamics transition from negative to positive as x increases, identifying x^* as an unstable repeller. We first observe that the total submission volume $V(x)$ is linearly decreasing in x, given the physical constraint $\alpha < 1$:

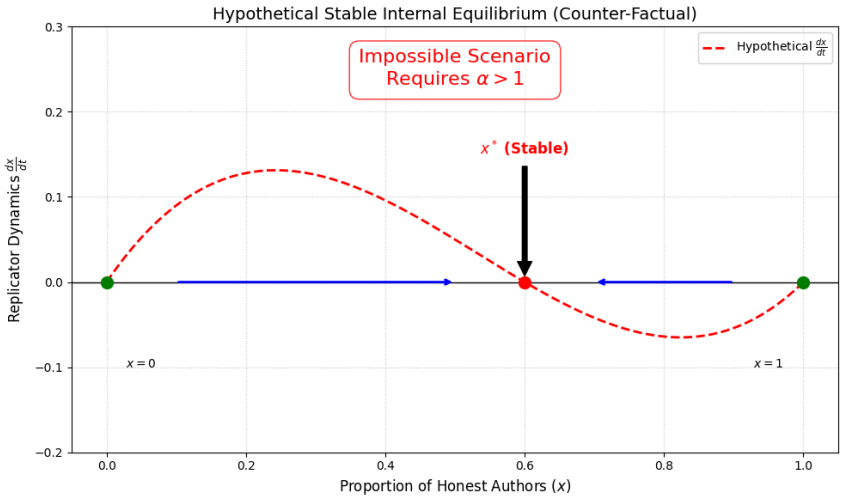


Figure 4: Hypothetical representation of a stable internal equilibrium (Counter-Factual). This dynamic profile, where $\dot{x}$ transitions from positive to negative at x^*, would require the strictly decreasing volume function $V(x)$ to be increasing. Mathematically, this implies $\alpha > 1$, which violates the physical definition of probability. Thus, a stable mixed strategy is structurally impossible in this model.

$$\frac{dV(x)}{dx} = \frac{d}{dx}\left[N(1-(1-\alpha)x)\right] = -N(1-\alpha) < 0 \tag{43}$$

Since the screening probability $\sigma(x)$ is inversely proportional to the volume $V(x)$, and the indicator function $\Psi(x)$ is linear in $\sigma(x)$, we apply the chain rule to determine the monotonicity of the system:

$$\frac{d\Psi}{dx} = \underbrace{\frac{\partial \Psi}{\partial \sigma}}_{(D+\lambda B)>0} \cdot \underbrace{\frac{d\sigma}{dV}}_{-\frac{K}{C_J V^2}<0} \cdot \underbrace{\frac{dV}{dx}}_{-N(1-\alpha)<0} \tag{44}$$

The product of two negative terms and one positive term yields a strictly positive result: $\Psi'(x) > 0$ for all $x \in [0, 1]$. This strict monotonicity proves that the function $\Psi(x)$ can cross zero at most once, and must do so from below. Consequently, the replicator dynamics $\dot{x}$ must change sign from negative to positive at x^*. This mathematically precludes the existence of a stable internal equilibrium (which would require $\Psi'(x) < 0$) and confirms that the system exhibits global bistability, where x^* acts as a basin boundary separating the regions of convergence to full corruption ($x = 0$) and full honesty ($x = 1$). We performed a parameter sweep to analyze how the tipping point x^* shifts with N and K.

4 Parameter Sensitivity and Bifurcation Analysis

In this section, we explore how the internal equilibrium x^* shifts in response to variations in the journal capacity K, which is the primary determinant of the congestion effect in our model. We analyze the conditions under which the system transitions between the three identified regimes: Global Corruption ($x^* \to 1$), Bistability ($0 < x^* < 1$), and Global Honesty ($x^* \to 0$). Unless otherwise stated, the numerical analysis assumes the default parameter set specified in Table 3. Note that the default capacity is set to $K = 100$, which corresponds to the bistable (congested) scenario observed in the previous analysis.

4.1 Impact of Journal Capacity K

The journal capacity K serves as the fundamental supply-side constraint of the peer-review system, determining the maximum number of submissions that can be rigorously screened per unit time. We

Table 3: Default Parameters for Numerical Analysis

Symbol	Description	Value
N	Total population of authors	1000
K	Journal Capacity	100
α	Probability of producing a good paper	0.2
ϵ	False negative rate	0.01
λ	Return rate for bad papers (False positive)	0.8
B	Benefit of acceptance	100
D	Penalty for accepted bad paper	50
C	Cost of submission	10
C_J	Unit cost of review	0.5

analyze the sensitivity of the internal equilibrium x^* to variations in capacity while holding all other parameters constant. The relationship is derived by equating the volume of submissions generated by the population to the maximum volume the journal can support at the critical screening intensity σ^*. Mathematically, the equilibrium requires the generated volume $V(x^*)$ to match the capacity-constrained limit:

$$\underbrace{N[1-(1-\alpha)x^*]}_{\text{Submission Volume Generated}} = \underbrace{\frac{1}{C_J}\left(\frac{K}{\sigma^*}-\epsilon\right)}_{\text{Capacity-Allowed Volume}} \tag{45}$$

Solving this linear equation for x^*, we obtain the explicit analytical dependence of the internal equilibrium on capacity:

$$x^*(K) = \frac{1}{1-\alpha}\left[1-\frac{1}{NC_J}\left(\frac{K}{\sigma^*}-\epsilon\right)\right] \tag{46}$$

where $\sigma^* = \frac{\lambda B - C}{D+\lambda B}$. This derived function reveals a strictly decreasing linear trend: as capacity K increases, the system can tolerate a higher submission volume, thereby reducing the requirement for honest authors (lowering x^*). This dynamic is visualized in Figure 5, which clearly delineates three distinct stability regimes based on the availability of screening resources.

The transition of the academic community through these regimes is driven by the interplay between resource supply and congestion demand. In the regime of Global Corruption, observed when capacity is critically low ($K < 53.9$ under default parameters), the journal is severely under-resourced. Even under the most favorable scenario where the entire population is honest ($x = 1$) and generates only the minimum submission volume $V_{min} = N\alpha$, the available capacity remains insufficient to maintain the critical screening probability σ^*. Mathematically expressed as $x^* > 1$, this condition implies that the system is structurally doomed; the journal lacks the physical capability to deter cheating regardless of the community's state, making universal corruption the inevitable evolutionary outcome. As capacity increases to an intermediate level ($53.9 \leq K \leq 269.2$), the system enters the Bistability Region, a phase where stability becomes conditional and path-dependent. Here, the journal possesses sufficient resources to screen a moderate volume of submissions but cannot sustain the maximum possible load. Consequently, the outcome depends on the initial state of the community: if the population is largely honest (high x), the submission volume remains low enough for effective screening, thereby reinforcing honest behavior. Conversely, if the community is initially corrupt (low x), the resulting flood of opportunistic submissions overwhelms the limited capacity, diluting screening effectiveness and creating a feedback loop that further incentivizes cheating. In this regime, the internal equilibrium $0 \leq x^* \leq 1$ acts as a tipping point separating these two distinct basins of attraction. Finally, when capacity becomes abundant ($K > 269.2$), the system achieves a state of Global Honesty. In this scenario, the journal possesses sufficient resources to rigorously screen the maximum possible submission volume ($V_{max} = N$) while maintaining the probability $\sigma \geq \sigma^*$. Because the screening threat remains credible regardless of the number of authors choosing to cheat, the internal equilibrium x^* drops below zero, signifying that the unstable tipping point has vanished from the physical state space. Under these conditions, the evolutionary dynamics unconditionally drive the population towards full honesty, as the high probability of detection renders the "Always Submit" strategy unprofitable under any circumstance, irrespective of the initial population distribution.

While the previous analysis of x^* describes the *intent* of the authors, Figure 6 illustrates the resulting *outcome* for the journal: the Average Publication Quality Q. The relationship reveals a critical distinction between "mechanical screening" and "strategic deterrence."

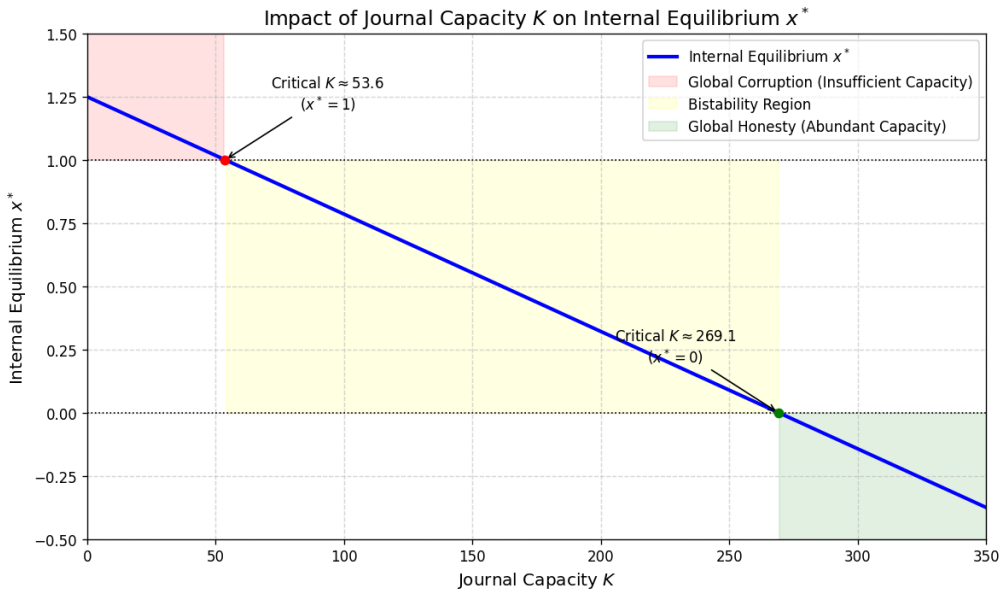


Figure 5: Impact of journal capacity K on the internal equilibrium x^*. The relationship is linearly decreasing. In the low capacity regime (Red), resources are insufficient to screen even minimal volumes, leading to Global Corruption ($x^* > 1$). As capacity increases, the system enters the Bistability region (Yellow). Finally, when capacity is abundant (Green), the journal can effectively screen all potential submissions, ensuring Global Honesty ($x^* < 0$).

In the **Corruption Regime** ($K < 269.2$), the system is trapped in a state where cheating remains profitable ($x^* > 0$, effectively $x = 0$ under worst-case assumptions). As capacity K increases within this region, we observe a gradual, linear improvement in quality (from $Q \approx 0.2$ to 0.35). This rise is purely mechanical: the journal has more resources to catch bad papers, increasing the realized screening probability σ. However, because the incentive structure has not fundamentally changed, authors continue to submit bad papers en masse. The journal is fighting a war of attrition—improving its filter but failing to stop the flood of submissions.

The most striking feature is the **Phase Transition** at the critical threshold $K \approx 269.2$. At this precise point, the capacity becomes sufficient to enforce a screening probability σ^* that neutralizes the expected utility of cheating. The vertical jump in quality from ≈ 0.35 to 1.0 represents a sudden strategic shift: rational authors abruptly cease submitting bad papers, not because they are all caught, but because the *risk* of detection has finally outweighed the *reward*.

This creates a "Perception Cliff": below the threshold, investing in capacity yields only diminishing marginal returns in quality; crossing the threshold yields an instantaneous recovery of the journal's reputation. This finding suggests that a journal cannot gradually reform itself out of corruption; it must secure enough capacity to trigger the phase transition.

4.2 Impact of Total Population N

While the journal capacity K represents the supply side of the screening resources, the total population N determines the demand intensity. In this section, we analyze how the internal equilibrium x^* shifts as the population size varies, assuming a fixed capacity (default $K = 100$). The relationship between population size and the tipping point is governed by the congestion mechanism. Analytically, the internal equilibrium x^* is the state where the incentive to cheat is exactly neutralized ($E[U_{Bad}] = 0$). This requires a specific screening probability $\sigma^* = \frac{\lambda B - C}{D + \lambda B}$. By substituting the definition of σ into the volume function $V(x) = N[1 - (1 - \alpha)x]$, we can solve for x^* as a function of N:

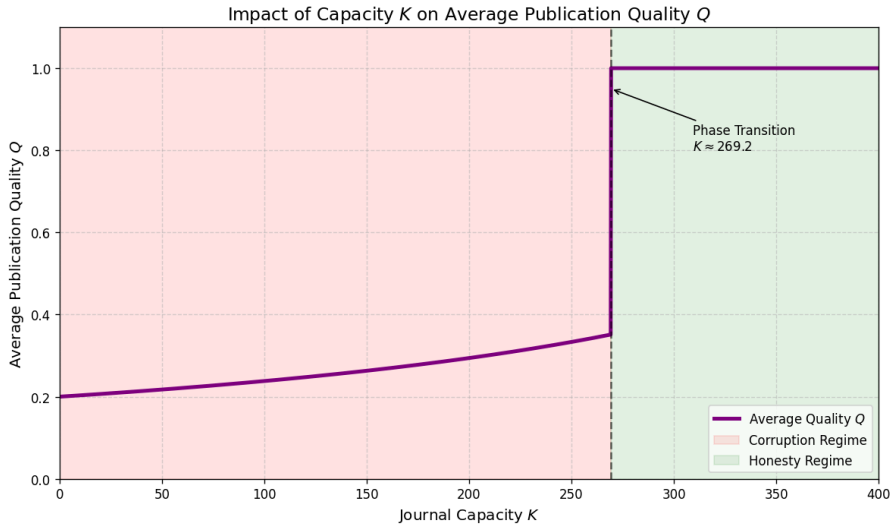


Figure 6: The impact of Journal Capacity K on the Average Publication Quality Q. The system exhibits a discontinuous phase transition at the critical capacity $K \approx 269.2$. In the Corruption Regime (Red), quality improves marginally due to mechanical screening but remains low. Once capacity exceeds the threshold, the system jumps to the Honesty Regime (Green), achieving perfect quality ($Q = 1.0$).

$$x^*(N) = \frac{1}{1-\alpha}\left(1 - \frac{1}{N \cdot C_J}\left[\frac{K}{\sigma^*} - \epsilon\right]\right) \tag{47}$$

This equation reveals a non-linear relationship between population size and system stability, as illustrated in Figure 7. We identify three distinct regimes:

In the Global Honesty Regime, observed when the population N is small (corresponding to the left side of the curve), the calculated tipping point x^* becomes negative. Mathematically, this implies that for any physically valid state $x \in [0, 1]$, the system positions itself to the "right" of the unstable repeller. Consequently, the evolutionary dynamics unconditionally drive the system towards full honesty ($x = 1$). Physically, this corresponds to a scenario where the population is sufficiently small that the journal maintains the capacity to effectively screen all submissions even under the worst-case submission loads, rendering cheating consistently unprofitable. Conversely, in the Global Corruption Regime, which occurs when N becomes extremely large (the right side of the curve), the tipping point x^* exceeds unity. In this situation, the entire physical range $[0, 1]$ lies to the "left" of the repeller, inevitably causing the system to collapse towards full corruption ($x = 0$). This collapse arises because the sheer volume of submissions generated by such a large population overwhelms the fixed capacity, rendering screening ineffective regardless of the collective behavior of the authors. For intermediate populations (highlighted in yellow), the system enters the Bistability Regime, where the unstable equilibrium x^* lies strictly within the physical range. This phenomenon indicates that the long-term outcome is not determined solely by system parameters but is instead *path-dependent*. If the initial proportion of honest authors is high ($x_0 > x^*$), the community generates enough "self-discipline" to keep submission volumes low, allowing the journal to maintain effective screening and causing the system to converge to honesty ($x = 1$). However, if the initial honest proportion is low ($x_0 < x^*$), the high volume of junk submissions dilutes the screening capacity, making cheating profitable and driving the system towards corruption ($x = 0$). Bistability represents a "trap": a journal that has lost its reputation (low x) cannot recover simply by maintaining the same resources that would sustain a healthy journal. Recovery would require a massive reduction in N or increase in K to shift x^* back to zero.

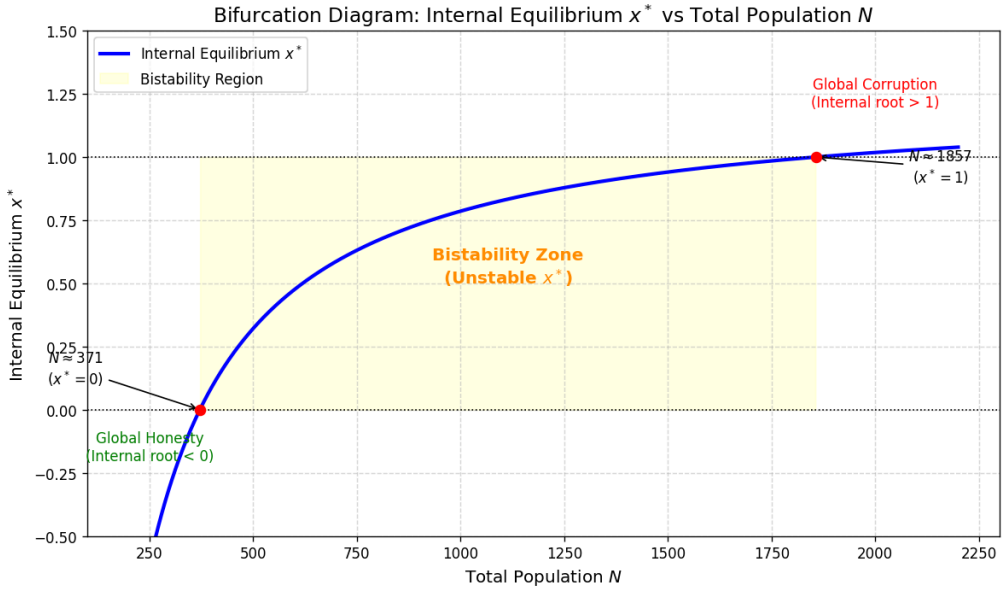


Figure 7: Bifurcation diagram showing the dependence of the internal equilibrium x^* on the total population N. The yellow shaded region indicates the zone of **Bistability**, where the system's fate depends on initial conditions. The critical thresholds mark the transitions to Global Honesty ($N \lesssim 371$) and Global Corruption ($N \gtrsim 1857$).

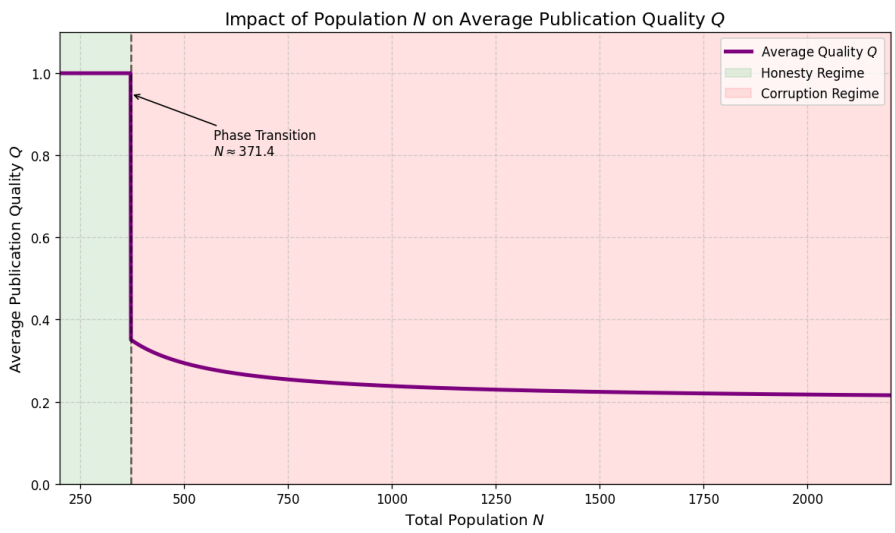


Figure 8: The impact of Total Population N on the Average Publication Quality Q. Similar to the capacity analysis, the system undergoes a sharp phase transition. When the population exceeds the critical carrying capacity ($N \approx 371.4$), the submission volume overwhelms the journal's fixed resources ($K = 100$), causing quality to plummet from 1.0 to ≈ 0.35.

While the bifurcation analysis in Figure 5 outlines the theoretical stability regions, Figure 8 illustrates the catastrophic consequence for the journal's output quality.

The Symmetry of Collapse: By comparing Figure 8 with the previous capacity analysis (Figure 6), we identify a fundamental symmetry. In the capacity analysis, the system failed when resources dropped below a critical threshold ($K \approx 269.2$ for $N = 1000$); here, the system fails when the load exceeds a critical limit ($N \approx 371.4$ for $K = 100$).

This confirms that the ecosystem's integrity is not determined by N or K in isolation, but by the *ratio* of screening capacity to submission volume. The "Phase Transition" at $N \approx 371.4$ represents the precise moment where the population's demand saturates the journal's screening capability. Beyond this point, the realized screening probability σ falls below the deterrence threshold σ^*. Consequently, the strategy of "Strategic Deterrence" collapses, and the journal enters a "Congestion Trap": the flood of bad papers (generated by $N > N_{crit}$) further dilutes the screening effectiveness, locking the system into a low-quality equilibrium ($Q \approx 0.35$) that cannot be reversed simply by reducing the population marginally.

4.3 Impact of Author Productivity α

Finally, we investigate the influence of the author's productivity parameter α, which represents the probability that a submission is genuinely "Good". While one might intuitively expect higher productivity to benefit the system, our analysis reveals a counter-intuitive dynamic under capacity constraints. Holding the population $N = 1000$ and capacity $K = 100$ constant, the stability of the system is determined by whether the submission volume exceeds the critical threshold V_{crit} required to maintain effective screening. The relationship between the internal equilibrium x^* and α is derived by equating the actual submission volume to this critical limit:

$$V(x^*) = N[1 - (1 - \alpha)x^*] = V_{crit} \tag{48}$$

where the critical volume is determined by the screening parameters as $V_{crit} = \frac{1}{C_J}(\frac{K}{\sigma^*} - \epsilon) \approx 371.4$. Solving this equation for x^*, we obtain a hyperbolic relationship that characterizes the tipping point:

$$x^*(\alpha) = \frac{1 - \frac{V_{crit}}{N}}{1 - \alpha} \tag{49}$$

The resulting bifurcation diagram, presented in Figure 9, illustrates how the stability of the academic ecosystem is sensitive to the capabilities of its participants. A striking feature of this relationship is that the equilibrium curve originates at a non-zero baseline of $x^* \approx 0.63$ when $\alpha = 0$. This intercept, which we term the system's "Base Load" of Honesty, reveals the fundamental structural constraints of the journal. In a scenario where honest authors produce no good papers ($\alpha = 0$), they effectively adopt a "Silence" strategy, submitting nothing. Meanwhile, opportunistic authors continue to saturate the journal with bad papers. To prevent the fixed capacity $K = 100$ from being overwhelmed, the number of active submitters must be strictly limited. Mathematically, the proportion of authors who must remain silent (honest) is governed by the ratio of critical capacity to total population:

$$x^*_{min} = x^*(0) = 1 - \frac{N_{AS}^{max}}{N} \approx 1 - \frac{371.4}{1000} \approx 0.629 \tag{50}$$

This implies that even in the absence of valid scientific output, a significant majority of the population must abstain from spamming for the peer-review system to function. As the productivity α increases, honest authors begin to contribute valid submissions to the pool, leading to a phenomenon we identify as the "Productivity Paradox". Paradoxically, the legitimate volume generated by productive honest authors adds to the overall congestion. To maintain the total volume at the sustainable level V_{crit}, the system must further compensate by reducing the number of opportunistic spammers. This necessitates an even higher proportion of honest authors to displace the noise with quality. Once α exceeds a critical threshold (≈ 0.37), the required honest fraction x^* exceeds unity:

$$x^*(\alpha) > 1 \quad \text{for} \quad \alpha > 1 - \frac{V_{crit}}{N} \tag{51}$$

At this point, the combined volume of high-quality submissions and opportunistic spam overwhelms the journal regardless of the strategy distribution, rendering the bistable equilibrium physically impossible and forcing the system into a state of Global Corruption.

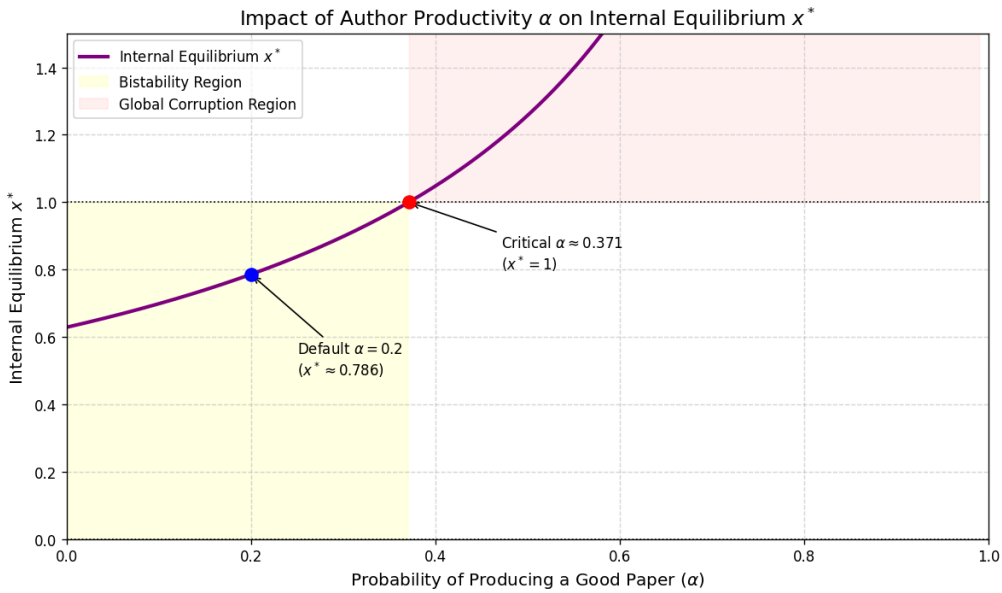


Figure 9: Dependence of the internal equilibrium x^* on author productivity α. The curve starts at a non-zero baseline of $x^* \approx 0.63$ at $\alpha = 0$. As productivity increases, the required proportion of honest authors rises until it exceeds physical bounds ($x^* > 1$) at $\alpha \approx 0.37$, triggering a transition from Bistability (Yellow) to Global Corruption (Red).

Figure 10: The impact of Author Productivity α on Average Publication Quality Q. Unlike the capacity and population metrics, quality Q exhibits a monotonic increase, creating a deceptive "Productivity Paradox." The system lacks a Global Honesty Regime (Green) due to initial saturation. Crucially, in the Global Corruption Regime (Red), Q continues to rise mechanically towards 1.0, masking the collapse of the screening system.

While Figure 8 demonstrates that higher productivity (α) destabilizes the strategic equilibrium x^*, Figure 10 reveals the paradoxical outcome for the journal's publication quality. This visualization highlights a dangerous "Trap of Prosperity" that is distinct from the dynamics observed in N and K.

Crucially, even after the system crosses the Sustainability Limit ($\alpha \approx 0.371$) and collapses into the Global Corruption Regime, the quality metric continues to improve mechanically. This improvement occurs because as the proportion of good papers in the submission pool increases, the final accepted set naturally contains a higher ratio of good papers, even if the journal's screening capability has been rendered ineffective by congestion. The rise in quality is therefore a reflection of the input material rather than the efficacy of the peer review process.

This dynamic creates a profound systemic risk best described as a "Trap of Prosperity." An observer looking solely at the output metric would see a thriving journal with improving standards approaching perfect quality, yet the underlying structure has transitioned into a state where the screening mechanism has failed and the journal is functionally defenseless against opportunism. This high-quality corruption represents a stable but fragile state where the journal's reputation is sustained not by its rigorous process, but merely by the high productivity of its authors, leaving it vulnerable to any sudden shift in submission behavior.

4.4 Impact of Screening Friction ϵ

The parameter ϵ in the screening function $\sigma(x) = \frac{K}{C_J V(x)+\epsilon}$ represents the operational friction or fixed overhead of the peer-review process that is independent of the submission volume. This could include administrative bottlenecks, platform latency, or the minimum time required to process a paper regardless of its quality. We analyze the sensitivity of the internal equilibrium x^* to ϵ by substituting the variable ϵ into the equilibrium condition $V(x^*) = \frac{1}{C_J}(\frac{K}{\sigma^*} - \epsilon)$. Solving for x^*, we observe a linear relationship:

$$x^*(\epsilon) = \frac{1}{1-\alpha}\left[1 - \frac{1}{NC_J}\left(\frac{K}{\sigma^*} - \epsilon\right)\right] \tag{52}$$

Taking the derivative with respect to ϵ, we find:

$$\frac{dx^*}{d\epsilon} = \frac{1}{(1-\alpha)NC_J} > 0 \tag{53}$$

This positive derivative indicates that an increase in system friction strictly increases the threshold for stability. The dynamic response is illustrated in Figure 11.

Physical Interpretation of Extremes

1. **Ideal Frictionless State ($\epsilon \to 0$)**: In this theoretical limit, the journal suffers no fixed overhead. All capacity K is dedicated strictly to the variable cost of reviewing submissions. This scenario yields the minimum possible value for x^*, representing the most robust state of the system. Even here, however, stability is not guaranteed if N is sufficiently large, as the variable cost $C_J V(x)$ alone can saturate capacity.

2. **High Friction State ($\epsilon \to 1$)**: When ϵ is large, a significant portion of the journal's "effective capacity" is consumed by overhead before any actual review takes place. To satisfy the critical screening condition $\sigma = \sigma^*$, the term $C_J V(x)$ must be reduced to compensate for the large ϵ. Physically, this means the community must drastically reduce its submission volume to accommodate the system's inefficiency. Since only honest authors reduce volume (by withholding bad papers), the system demands a much higher proportion of honest participants (higher x^*) to survive. If ϵ becomes too large, x^* may exceed 1, pushing the system into inevitable Global Corruption solely due to administrative inefficiency.

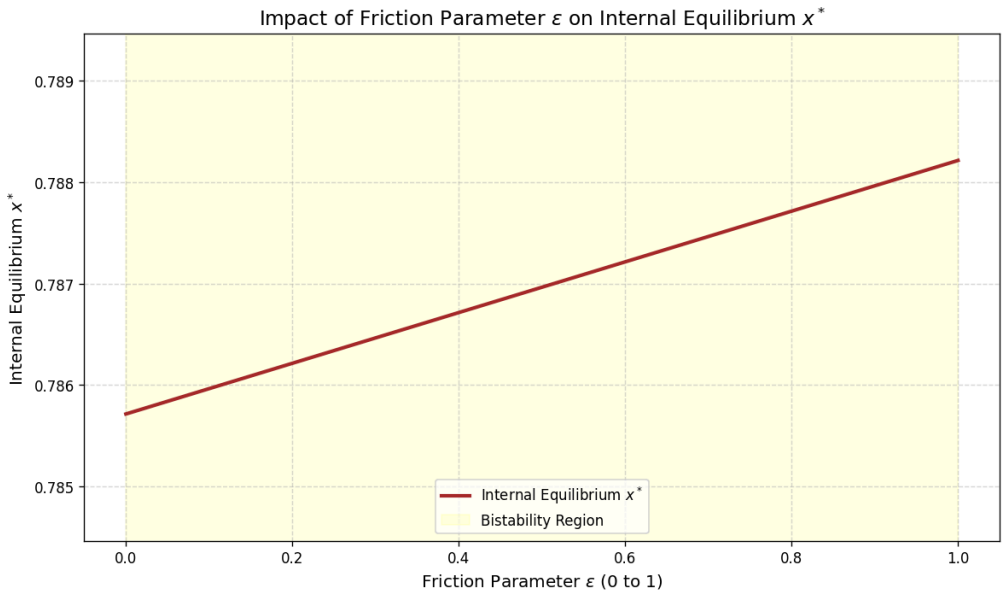


Figure 11: Linear dependence of the internal equilibrium x^* on the screening friction ϵ. An increase in friction consumes journal resources, necessitating a higher proportion of honest authors (higher x^*) to reduce volume and maintain screening effectiveness.

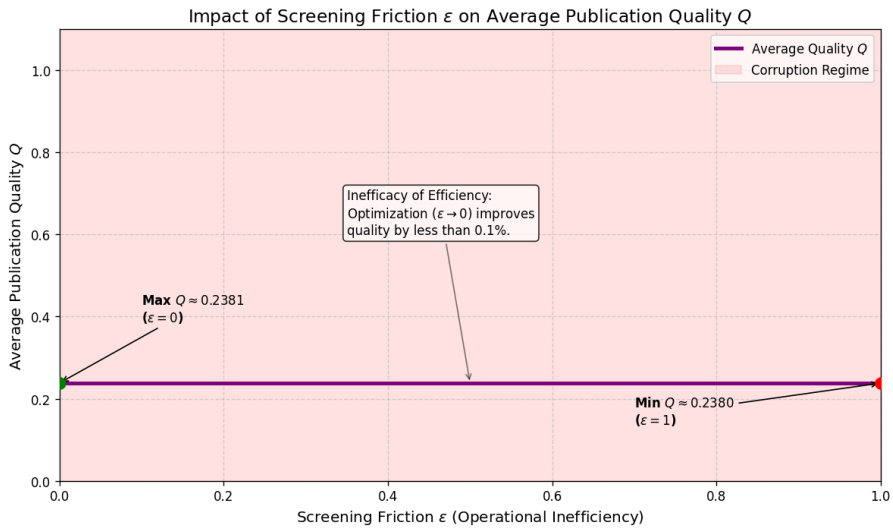


Figure 12: The impact of Screening Friction ϵ on Average Publication Quality Q. The flat trajectory reveals the limits of operational efficiency. Under the default high-load conditions ($N = 1000$), the system remains trapped in the Corruption Regime (Red) across the entire range of ϵ. Reducing friction to zero yields negligible quality improvements because the variable cost of reviewing the massive volume of submissions ($C_J \cdot N$) already exceeds the journal's capacity.

Complementing the stability analysis, Figure 12 illustrates the insensitivity of publication quality Q to the friction parameter ϵ. This result provides a critical "sanity check" for policy interventions focused solely on administrative optimization.

The curve is virtually flat. Numerical analysis reveals that reducing friction from its maximum ($\epsilon = 1.0$) to perfect efficiency ($\epsilon = 0.0$) yields a negligible improvement, raising quality from a minimum of $Q \approx 0.3857$ to a maximum of only $Q \approx 0.3861$. This gain of less than 0.1% indicates that the "bottleneck" of the system is not the fixed overhead (ϵ), but the variable load generated by the population (N). When the journal is structurally overwhelmed (where $N \cdot C_J \gg K$), optimizing the submission system or reducing administrative delays ($\epsilon \to 0$) provides no strategic benefit. It is akin to widening the door of a room that is already full; it does not change the capacity constraint.

The absence of a phase transition in this plot confirms that the system is structurally locked into the Corruption Regime. The condition for the Honesty Regime ($x^* < 0$) would require $\epsilon < K/\sigma^* - NC_J$. Under default parameters, this threshold is negative (≈ -75), which is physically impossible. This mathematical impossibility signifies that no amount of administrative streamlining can restore the journal's reputation without fundamentally addressing the primary imbalances in Capacity (K) or Population (N).

4.5 Impact of False Positive Reward λ

Finally, we investigate the role of λ, the false positive rate, which serves as a proxy for the return rate on bad papers. Unlike capacity parameters (K, ϵ), λ directly alters the payoff structure of the game. Our analysis highlights that the influence of λ cannot be evaluated in isolation; rather, it must be synthesized with the benefit B and submission cost C to determine the *Profitability Condition* of the "Always Submit" strategy. The system's behavior is fundamentally governed by the net expected profit of a successful speculative submission, given by $\pi_{bad} = \lambda B - C$. We identify three distinct phases of stability based on this net profit, as illustrated in Figure 13:

1. **Phase I: Unconditional Honesty ($\lambda B \leq C$)**: In the region where $\lambda \leq C/B$ (approximated as $\lambda \leq 0.1$ under default parameters), the expected reward from a bad paper, even if accepted, is insufficient to cover the submission cost. In this regime, the strategy "Always Submit" is strictly dominated by "Only Good" or "Never Submit" regardless of the journal's screening effectiveness. The internal equilibrium x^* is mathematically negative, implying that the system converges to Global Honesty unconditionally.

2. **Phase II: Bistability and the Rising Threshold ($0 \leq x^* \leq 1$)**: Once λ crosses the profitability threshold, cheating becomes economically viable, and the system enters the Bistable regime. We observe a monotonic increase in x^* as λ grows. The mechanism driving this trend is the *Incentive-Congestion Trade-off*. As λ increases, the incentive to cheat rises. To neutralize this higher incentive ($E[U_{Bad}] = 0$), the journal must impose a strictly higher screening probability σ^*:

$$\sigma^*(\lambda) = \frac{\lambda B - C}{D + \lambda B} \tag{54}$$

   Since $\frac{\partial \sigma^*}{\partial \lambda} > 0$, a higher λ demands stricter screening. Under fixed capacity K, increasing screening efficiency requires a reduction in submission volume $V(x)$. Consequently, the population must become increasingly honest (higher x^*) to sufficiently lower the volume and "afford" the required screening intensity.

3. **Phase III: Global Corruption ($x^* > 1$)**: When λ exceeds a critical upper bound ($\lambda \approx 0.6$), the temptation to cheat becomes so high that the required screening probability σ^* demands a submission volume lower than physically possible (even if $x = 1$). Mathematically, x^* crosses unity. At this point, even a fully honest population generates too much volume to sustain the screening intensity needed to deter the high-reward cheating, causing the system to collapse into Global Corruption.

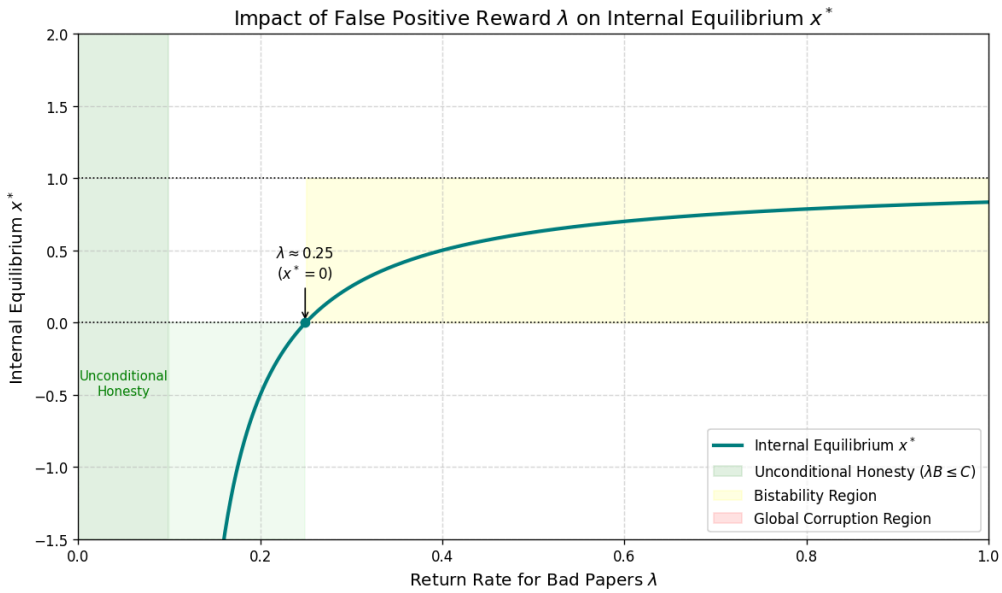


Figure 13: Impact of the false positive return rate λ on the internal equilibrium x^*. The system exhibits Unconditional Honesty (Green) when cheating is unprofitable ($\lambda B \leq C$). As λ increases, the profitable region requires an increasing proportion of honest authors to maintain stability (Yellow), eventually leading to inevitable collapse (Red) when the incentive exceeds the system's maximum screening potential.

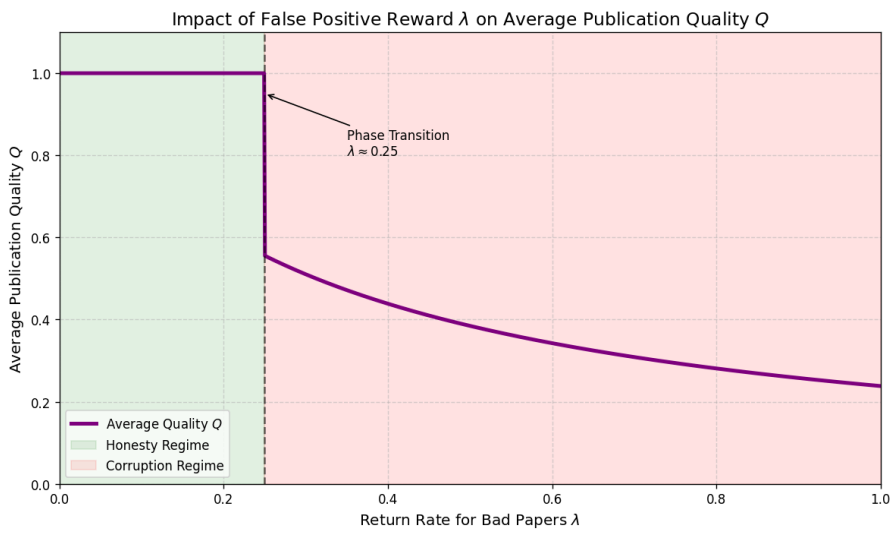


Figure 14: The impact of False Positive Reward λ on Average Publication Quality Q. The system exhibits a sharp phase transition at $\lambda \approx 0.25$. Unlike the operational friction ϵ, the strategic parameter λ acts as a "switch" for corruption. Once the return on bad papers exceeds the critical threshold, quality collapses immediately and continues to degrade as λ increases, directly diluting the publication pool.

While the previous stability analysis focused on the intent to cheat (x^*), Figure 14 illustrates the catastrophic consequence of the false positive rate λ on the actual publication quality Q. The graph reveals a brutal phase transition at $\lambda \approx 0.25$. Below this threshold, the system resides in the Honesty Regime where quality is perfect ($Q = 1.0$), as the low probability of success renders cheating irrational. However, once λ crosses the critical point where the expected return covers the submission cost, the system does not degrade gradually; it collapses instantly. Following this phase transition, Q follows a declining curve in the Corruption Regime. This downward trajectory occurs because λ acts not only as an incentive but also as a direct "noise injector": as λ increases, a higher proportion of the bad papers submitted by the now-corrupt population physically penetrate the review filter, mathematically diluting the ratio of good papers in the final publication record.

This result stands in stark contrast to the negligible impact of screening friction ϵ observed earlier. While ϵ is merely an "administrative parameter" that affects the processing speed of the flood, λ functions as a "strategic parameter" that controls the source of the flood itself. In our model, λ enters directly into the author's utility function ($\pi_{bad} = \lambda B - C$), granting it the power to switch the community's strategy from "silence" to "spamming." The comparison between the flat trajectory of ϵ and the sharp cliff of λ highlights a fundamental asymmetry: administrative inefficiency (ϵ) is a bottleneck, but high false-positive rewards (λ) are a structural fatal flaw. No amount of administrative optimization (lowering ϵ) can save a journal if the strategic incentive structure (λ) makes fraud profitable.

The real-world implication of this dynamic is significant for academic policy. It suggests that the current focus on "optimizing peer review workflows" or "increasing submission speeds" (reducing ϵ) is largely futile in the face of structural congestion. To prevent the collapse of scientific quality, interventions must focus strictly on the "strategic" side of the equation: reducing the utility of bad papers. This means technically lowering λ (through stricter, albeit slower, review standards) or economically neutralizing it (by increasing penalties D). The model confirms that a slow but rigorous journal is sustainable, whereas a fast but permeable one is destined for immediate collapse into the low-quality trap.

4.6 Impact of Acceptance Benefit B

We further examine the influence of the reward for acceptance, denoted by B. This parameter represents the career advancement, funding, or prestige associated with a publication. The relationship between B and the internal equilibrium x^* is governed by the profitability condition of the "Always Submit" strategy. Analytically, the critical screening probability σ^* required to neutralize the incentive to cheat is given by:

$$\sigma^*(B) = \frac{\lambda B - C}{D + \lambda B} \tag{55}$$

Substituting this into the equilibrium solution for x^*, we obtain the response curve illustrated in Figure 15.

For low values of B, specifically when $\lambda B \leq C$, the expected payoff of a speculative submission is negative even in the absence of screening ($\sigma = 0$). In this **Unprofitable Zone**, rational opportunistic authors voluntarily abstain from submitting bad papers because the cost of submission C outweighs the probability-weighted benefit. Mathematically, the equilibrium condition requires $\sigma^* \leq 0$, which is physically realized as unconditional stability ($x^* \to -\infty$). Under default parameters, this threshold occurs at $B \leq 12.5$. A critical question arises regarding the system's behavior as the reward B becomes arbitrarily large. As shown in Figure **??**, x^* does not diverge to infinity but instead converges asymptotically to a value slightly greater than 1.

Mathematically, as $B \to \infty$, the penalty D becomes negligible relative to the reward, and the required screening probability approaches unity:

$$\lim_{B\to\infty} \sigma^*(B) = \lim_{B\to\infty} \frac{\lambda B - C}{D + \lambda B} = 1 \tag{56}$$

This implies that to stop authors from cheating for an infinite reward, the journal must catch 100% of bad papers. The critical submission volume that the journal can support at this maximum screening intensity is fixed by the physical capacity K:

$$V_{crit}^{\infty} = \lim_{\sigma\to 1} \frac{1}{C_J}\left(\frac{K}{\sigma^*} - \epsilon\right) = \frac{K - \epsilon}{C_J} \tag{57}$$

Under default parameters ($K = 100, \epsilon = 0.01, C_J = 0.5$), this limit is $V_{crit}^{\infty} = 199.98$. However, even if the entire population is honest ($x = 1$), the minimum "honest" submission volume is $V_{min} = N\alpha = 200$.

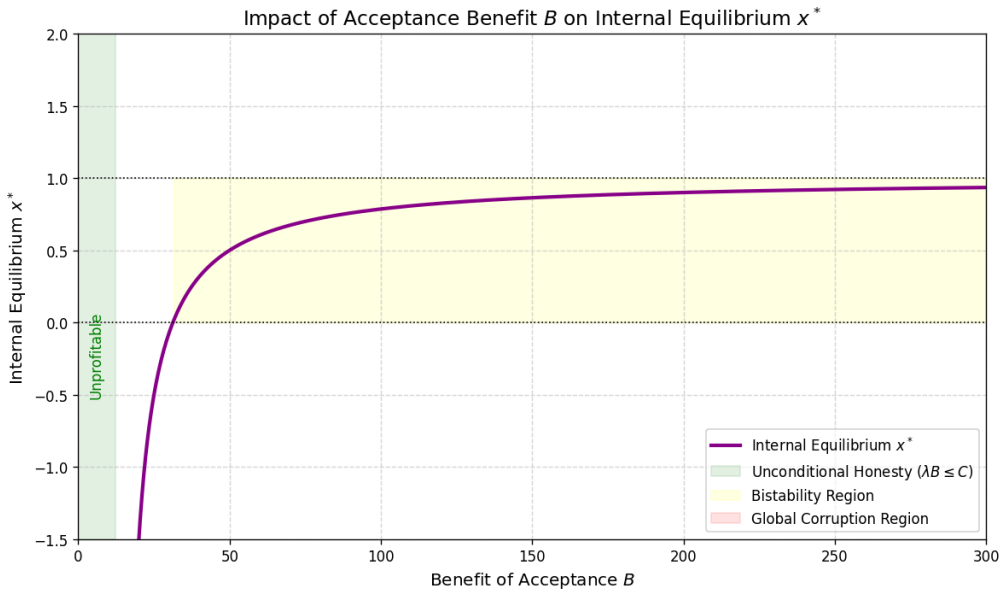


Figure 15: Impact of acceptance benefit B on the internal equilibrium x^*. The Green zone indicates the **Unprofitable Region** where the reward is insufficient to cover submission costs ($\lambda B \leq C$), leading to unconditional honesty. As B increases beyond this threshold, the system enters the Bistable regime (Yellow), requiring strict screening to offset the rising temptation.

Since $V_{min} > V_{crit}^{\infty}$, the system is structurally incapable of providing $\sigma = 1$ screening for the whole population. Consequently, the asymptote x_{inf}^* settles slightly above 1:

$$x_{inf}^* = \frac{1 - V_{crit}^{\infty}/N}{1 - \alpha} \approx 1.000025 \tag{58}$$

This result signifies that while high rewards drive the system towards corruption, the failure is driven by the **physical saturation** of capacity rather than the economic incentive itself. The journal simply cannot physically review even the honest papers strictly enough to deter the infinite temptation.

The relationship between the Acceptance Benefit B and publication quality Q reveals the decoupling of economic incentives from physical outcomes once a system is saturated. The phase transition at $B \approx 31.2$ marks the "Tipping Point of Greed," where the expected return on fraud surpasses the journal's deterrence threshold. Before this point, the deterrent (D) dominates, ensuring perfect quality ($Q = 1.0$). Immediately after crossing this threshold, the strategic equilibrium flips to Global Corruption.

Crucially, the quality curve in the Corruption Regime is a horizontal line. This flatness signifies a "Saturation of Failure." Once B is high enough to induce the "Always Submit" strategy ($x = 0$), the system reaches its physical limit: the submission volume hits the population cap (N), and the screening capability hits the capacity floor (K). Further increasing B—even to infinity—cannot induce "more" cheating, nor can it further degrade a screening process that is already overwhelmed. The economic driver (B) has done its damage, and the system is now locked into a state determined solely by physical constraints.

In this collapsed state, the average quality Q settles at approximately 0.2, which corresponds to the author productivity α. This occurs because the overwhelmed journal effectively loses its ability to filter, reducing the selection process to a near-random sampling of the submission pool. Since the pool contains a natural mix of α good papers and $(1 - \alpha)$ bad papers, the output quality mirrors the input ratio. Interestingly, the technical parameters ϵ (friction) and λ (false positive rate) exert no visible influence on this baseline. In a state of structural congestion, the primary source of error is no longer the microscopic technical failure of reviewers (λ), but the macroscopic breakdown of processing capacity.

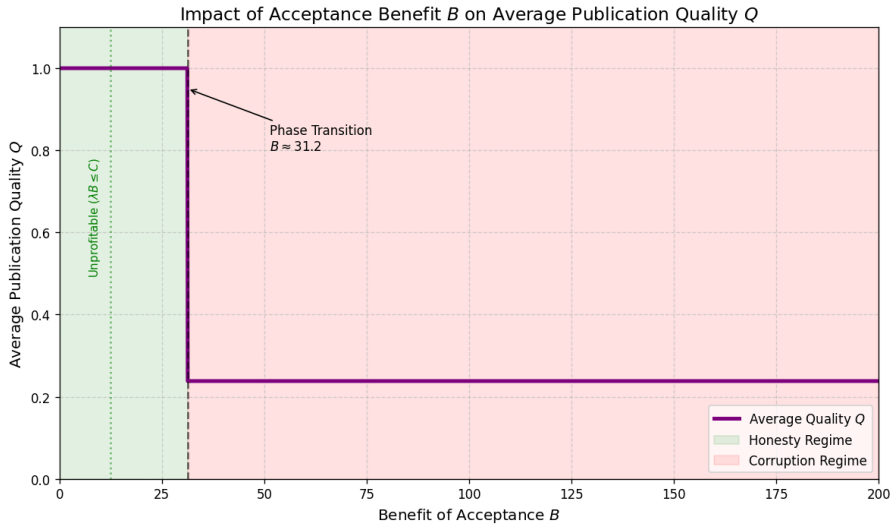


Figure 16: The impact of Acceptance Benefit B on Average Publication Quality Q. The system displays a binary outcome: perfect honesty or complete collapse, separated by a sharp phase transition at $B \approx 31.2$. The flat trajectory in the Corruption Regime indicates that once the incentive to cheat saturates the population's strategy, further increases in reward have no impact on the physical degradation of the journal.

The "flood" of submissions renders fine-tuned screening parameters irrelevant, anchoring the journal's quality firmly to the base productivity of the community.

4.7 Impact of Rejection Penalty D

The parameter D represents the reputational or administrative penalty imposed on an author when a submitted paper is identified as "Bad" and rejected. This serves as the primary deterrent mechanism in the ecosystem. Unlike the reward B, which fuels the incentive to cheat, the penalty D suppresses it. We analyze the sensitivity of the system to this deterrent by examining the critical screening probability σ^* required to maintain equilibrium ($E[U_{Bad}] = 0$). The required screening intensity is inversely related to the penalty magnitude:

$$\sigma^*(D) = \frac{\lambda B - C}{D + \lambda B} \tag{59}$$

As the penalty D increases, the denominator grows, reducing the necessary screening probability σ^*. Physically, this implies that a severe punishment allows the journal to be less rigorous in detection while still maintaining a sufficient deterrent effect. The relationship between the internal equilibrium x^* and D is governed by the capacity constraint equation:

$$x^*(D) = \frac{1}{1-\alpha}\left[1 - \frac{1}{NC_J}\left(\frac{K}{\sigma^*(D)} - \epsilon\right)\right] \tag{60}$$

Since x^* is a decreasing function of $1/\sigma^*$, and σ^* decreases with D, the internal equilibrium x^* is strictly decreasing with respect to D. The resulting dynamic landscape is visualized in Figure 17.

The curve in Figure 17 highlights three distinct regimes driven by the magnitude of the penalty. For low values of D, the risk of rejection is trivial compared to the potential reward. Consequently, the journal requires a near-perfect screening rate ($\sigma \to 1$) to neutralize the payoff, which demands an impossibly low submission volume, pushing the system into Global Corruption ($x^* > 1$). As D increases, the "risk premium" of cheating rises. The system enters the Bistable regime, where a moderate level of honesty is sufficient to maintain order. Finally, when D exceeds a critical threshold, the penalty becomes so severe that even a minimal screening probability (which allows for maximum submission

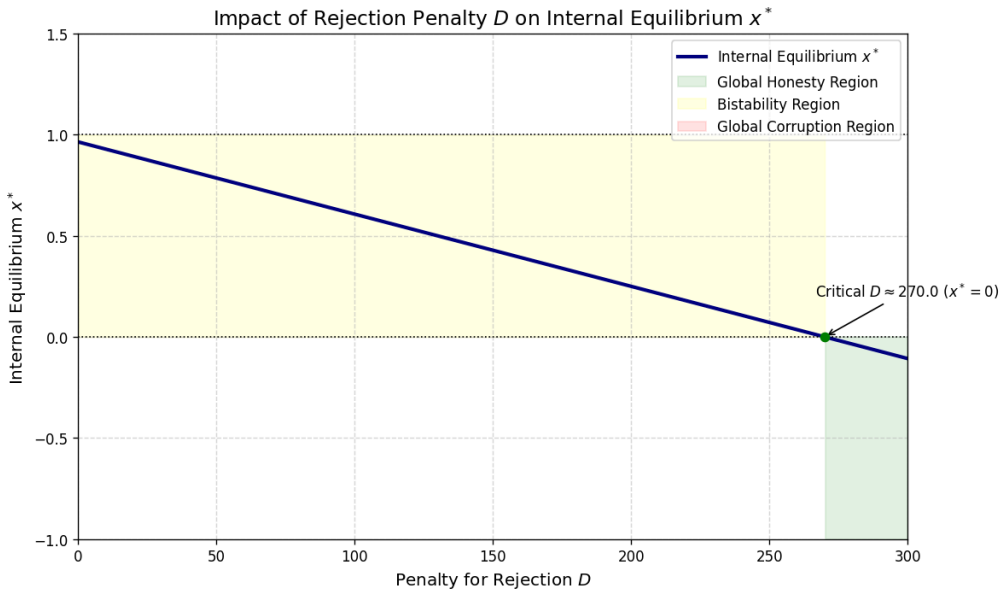


Figure 17: Impact of rejection penalty D on the internal equilibrium x^*. The relationship is monotonically decreasing. Low penalties (left) fail to deter cheating, requiring physically impossible screening levels ($x^* > 1$, Global Corruption). High penalties (right) create a strong deterrent even with relaxed screening, stabilizing the system at lower levels of honesty ($x^* < 0$, Global Honesty). The intermediate region (Yellow) represents the zone of Bistability.

volume) is sufficient to render cheating unprofitable. In this regime ($x^* < 0$), the system converges to Global Honesty regardless of the initial population state, as the cost of detection outweighs any potential gain.

Figure 18 completes the economic picture by illustrating the impact of the Rejection Penalty D on publication quality. The resulting curve is the precise geometric inverse of the Acceptance Benefit curve observed in Figure 16. While increasing the benefit B drives the system down a cliff into corruption, increasing the penalty D pushes the system up a step into honesty. The "Step-Up" phase transition at $D \approx 270.1$ represents the "Deterrence Threshold." Below this critical value, the penalty is insufficient to outweigh the immense expected gain from cheating ($\lambda B - C$). Consequently, the system remains saturated with bad papers, and the quality stays flat at the baseline level determined by author productivity ($\alpha \approx 0.2$). This flatness mirrors the behavior in the corruption regime of the Benefit analysis: once the penalty is too weak to deter the "Always Submit" strategy, making it *slightly* weaker (lowering D) does not worsen the quality, because the journal's physical capacity is already maxed out by the flood of spam.

The symmetry between the Benefit and Penalty curves carries a profound implication for institutional policy: Greed and Fear are not linear counterweights, but threshold-based switches. The "flatness" on the left side of the graph indicates that weak sanctions are functionally equivalent to no sanctions. In a congested system where the probability of detection is low (due to low σ), incremental increases in penalties—such as small fines or temporary suspensions—are completely futile if they remain below the critical threshold ($D < 270$). To restore a corrupted journal to the Honesty Regime, the administration must implement a "radical shock," increasing the penalty discontinuously to cross the threshold. This confirms that in a resource-constrained environment ($K \ll N$), where the journal cannot afford to screen everyone thoroughly, severe, high-stakes sanctions are the only viable mechanism to maintain scientific integrity.

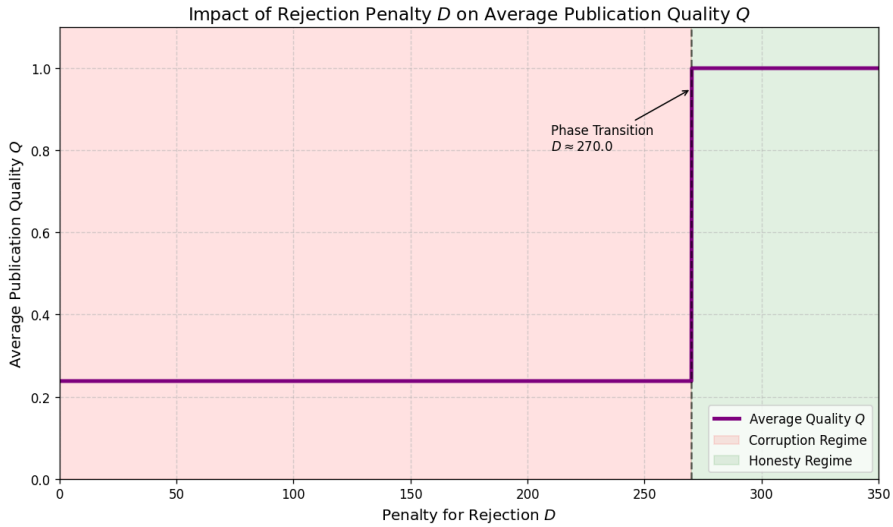


Figure 18: The impact of Rejection Penalty D on Average Publication Quality Q. The trajectory acts as the geometric inverse of the Benefit curve. The system is trapped in a flat, low-quality state ($Q \approx 0.24$) for all penalties below the critical threshold $D \approx 270.1$. Once the penalty exceeds this "Deterrence Threshold," the quality instantaneously jumps to perfection ($Q = 1.0$), confirming the non-linear nature of sanctions.

4.8 Impact of Submission Cost C

The submission cost C represents the economic or temporal barrier to entry for submitting a manuscript, including submission fees, formatting effort, or the opportunity cost of time. Unlike the capacity parameters which constrain the journal's throughput, C acts as a direct filter on the input side of the system by altering the net profitability of the "Always Submit" strategy, given by $\pi_{bad} = \lambda B - C$. The relationship between C and the internal equilibrium x^* is strictly monotonically decreasing. As the cost C rises, the net profit of a successful bad paper diminishes, allowing the journal to maintain effective deterrence with a lower screening probability σ^*. Mathematically, this relaxation is expressed as:

$$\sigma^*(C) = \frac{\lambda B - C}{D + \lambda B} \tag{61}$$

A lower required σ^* implies that the system can tolerate a higher volume of submissions while still maintaining equilibrium. With a higher allowable volume, the necessity for a large proportion of honest (silent) authors decreases. Thus, increasing C lowers the stability threshold x^*, making the system more robust against corruption. The bifurcation diagram visualized in Figure 19 illustrates the continuous transition across three distinct stability regimes driven by this cost structure.

At the lower bound where submission is nearly free ($C \to 0$), the profit margin for spamming is maximized. In this scenario, the journal is forced to maintain strict screening ($\sigma^* \to \frac{\lambda B}{D+\lambda B}$), which severely limits the allowable submission volume. If the total population N is large, even a fully honest population ($x = 1$) may generate excessive volume for such strict screening, resulting in $x^* > 1$ and driving the system into a Global Corruption regime. This finding suggests that "free-to-submit" policies in prestige journals are inherently unstable without massive review capacity. As the cost C increases, the burden on the screening system relaxes, and the equilibrium x^* falls into the physical range $(0, 1)$, creating a Bistability regime where the system's fate depends on initial conditions. Ultimately, a critical singularity emerges when the cost fully offsets the expected benefit of a false positive, specifically when $C \geq \lambda B$. In this Unconditional Honesty regime, the expected utility of a bad paper becomes negative even in the absence of screening ($\sigma = 0$). Consequently, rational opportunistic authors voluntarily abstain from submission, leading to $x^* \to -\infty$ and ensuring global stability regardless of journal capacity.

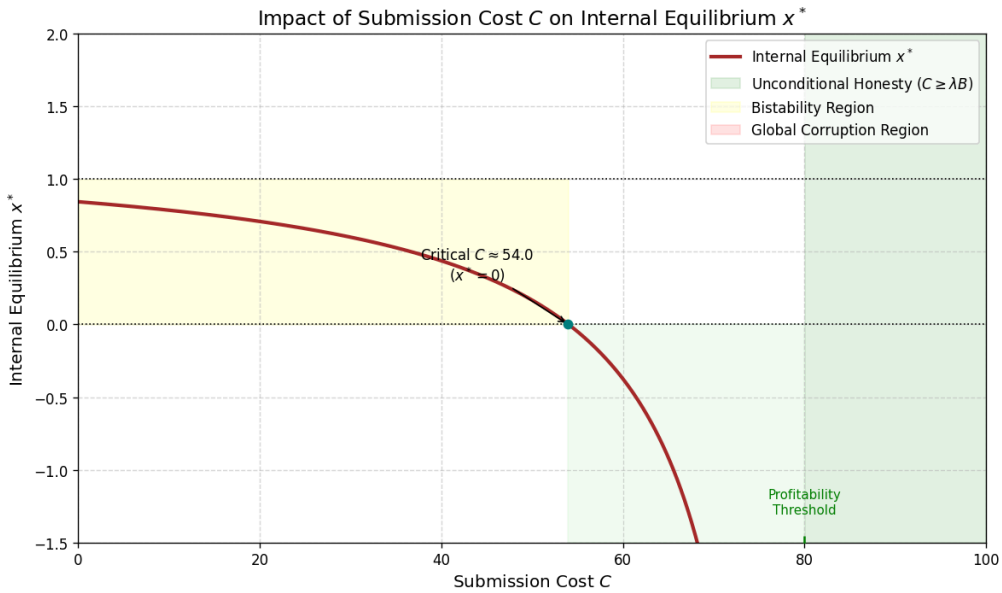


Figure 19: Impact of submission cost C on the internal equilibrium x^*. The relationship is monotonically decreasing. Extremely low costs (left) encourage spamming, pushing the system toward Global Corruption ($x^* > 1$). As costs rise, the system stabilizes. Once C exceeds the profitability threshold (λB), the system achieves Unconditional Honesty (Green) as cheating becomes economically irrational.

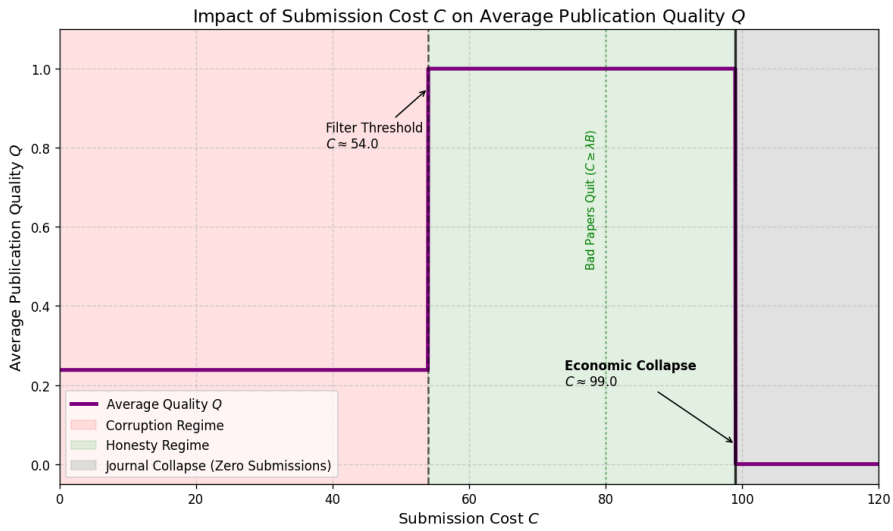


Figure 20: The impact of Submission Cost C on Average Publication Quality Q. The system demonstrates three distinct phases: (1) **Corruption Regime** (Red), where low costs ($C < 54.0$) invite spam; (2) **Honesty Regime** (Green), where moderate costs ($54.0 \leq C < 99.0$) filter out bad papers while retaining good ones; and (3) **Journal Collapse** (Gray), where excessive costs ($C \geq 99.0$) make submission unprofitable even for honest authors, resulting in zero submissions and the effective death of the journal.

The extended analysis of Submission Cost C in Figure 20 reveals a "Goldilocks Zone" for academic economics. The curve exhibits two critical transitions, framing submission cost as a double-edged sword. The first transition at $C \approx 54.0$, termed the "Filter Threshold," marks the point where the cost becomes sufficiently prohibitive to deter opportunistic spam. Below this threshold, in the Corruption Regime, the low barrier to entry invites a flood of bad papers, overwhelming the journal's capacity and trapping quality at the stagnant baseline ($Q \approx 0.24$) determined by the "Diluted Limit" of random sampling. Crossing this threshold triggers an instantaneous recovery to perfect quality ($Q = 1.0$), as the cost effectively filters out the noise while remaining affordable for valid research.

However, the corrected model reveals a second, fatal transition at $C \approx 99.0$, termed the "Collapse Threshold." If the cost continues to rise and exceeds the expected utility of a good paper ($B(1-\epsilon)$), the system enters the "Journal Collapse" zone. In this regime, the economic barrier is so high that it renders submission irrational even for honest authors with high-quality work. Consequently, the submission volume drops to zero, and the journal effectively dies. This stark drop in quality to zero (representing no output) serves as a critical warning: while economic barriers are essential to prevent congestion and corruption, excessive barriers—akin to exorbitant open-access fees—risk suffocating the scientific enterprise entirely by pricing out legitimate knowledge production.

This tri-phasic behavior highlights the delicate balance required in policy design. The "Saturation Plateau" on the left confirms that low-cost venues are structurally doomed to low quality due to congestion, validating the need for some friction. Yet, the "Collapse Zone" on the right warns against the unrestricted escalation of costs. A sustainable academic ecosystem must therefore operate strictly within the Honesty Regime ($54 < C < 99$), maintaining barriers high enough to deter fraud but low enough to foster discovery.

4.9 Impact of Unit Review Cost C_J

The parameter C_J represents the unit cost or resource consumption required to review a single submission. It serves as an inverse proxy for the journal's operational efficiency; a lower C_J implies a highly efficient workflow capable of processing large volumes, while a high C_J indicates a resource-intensive process. We analyze the sensitivity of the internal equilibrium x^* to this cost by rearranging the capacity constraint equation. For the system to maintain the screening probability σ^* required for equilibrium, the submission volume must satisfy $V(x^*) = \frac{1}{C_J}(\frac{K}{\sigma^*} - \epsilon)$. Substituting the volume definition $V(x) = N[1-(1-\alpha)x]$, we derive the relationship between stability and review cost:

$$x^*(C_J) = \frac{1}{1-\alpha}\left[1 - \frac{1}{NC_J}\left(\frac{K}{\sigma^*} - \epsilon\right)\right] \tag{62}$$

The trajectory of x^* as a function of C_J is illustrated in Figure 21. The curve exhibits a strictly increasing trend, indicating that as the review process becomes more expensive (or less efficient), the system requires a strictly higher proportion of honest authors to maintain stability. In the "High Efficiency" regime (low C_J), the journal can afford to screen a vast number of submissions. Consequently, the critical threshold x^* is low (or negative), meaning the system can remain honest even with a significant number of opportunistic submitters. However, as efficiency degrades (increasing C_J), the allowable volume contracts, forcing the ecosystem to rely increasingly on author self-restraint (honesty) to prevent congestion.

A critical feature of this dynamic is the existence of a horizontal asymptote as the cost becomes arbitrarily large, which we define as the **Saturation Limit**. As $C_J \to \infty$, the term representing the journal's processing capability approaches zero ($\frac{1}{C_J} \to 0$). In this limit, the equilibrium converges to:

$$\lim_{C_J \to \infty} x^*(C_J) = \frac{1}{1-\alpha} \tag{63}$$

Under the default parameter $\alpha = 0.2$, this limit is 1.25. The physical interpretation of this saturation limit is profound: if the review process becomes infinitely expensive, the journal's effective capacity drops to zero. At this point, even if the entire population is honest ($x = 1$) and submits only good papers, the resulting volume $V_{min} = N\alpha$ is still infinitely greater than the journal's zero capacity. Therefore, once the cost C_J exceeds the critical threshold where $x^* = 1$, the system is structurally doomed to Global Corruption, not because of spam, but because it lacks the efficiency to handle even the legitimate scientific output of the community.

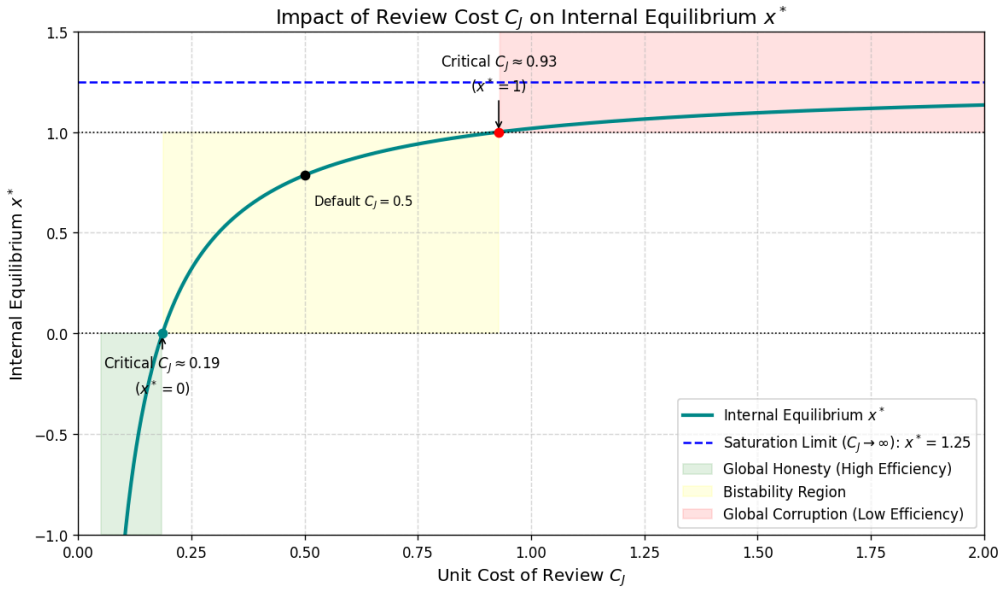


Figure 21: Impact of unit review cost C_J on the internal equilibrium x^*. The relationship is monotonically increasing. Highly efficient journals (Low C_J, Green) can sustain Global Honesty. As costs rise, the system enters Bistability (Yellow). In the limit of extreme inefficiency ($C_J \to \infty$), the curve approaches a **Saturation Limit** (Blue dashed line) determined solely by author productivity α.

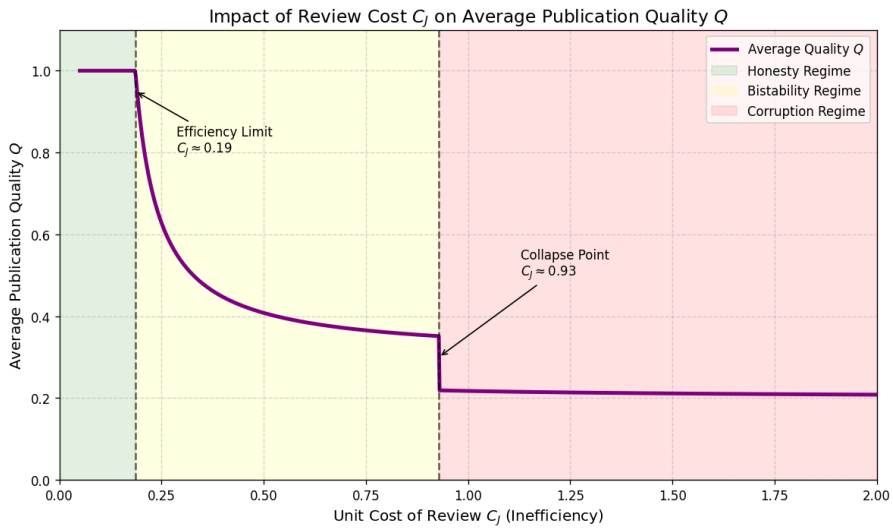


Figure 22: The impact of Unit Review Cost C_J on Average Publication Quality Q. The system behaves in reverse to capacity, with high costs (inefficiency) driving collapse. The graph reveals three distinct regimes: (1) **Honesty Regime** (Green), where high efficiency ($C_J < 0.23$) allows the journal to screen all submissions effectively; (2) **Bistability Regime** (Yellow), where rising costs ($0.23 < C_J < 1.16$) force the journal to ration capacity, allowing a gradual infiltration of bad papers; and (3) **Corruption Regime** (Red), where inefficiency ($C_J > 1.16$) renders screening impossible, locking quality at the random baseline ($Q \approx 0.24$).

The analysis of Unit Review Cost C_J in Figure 22 underscores that operational efficiency is not merely a budgetary concern but a structural determinant of scientific quality. The variable C_J represents the inverse of efficiency—the resources (time, money, effort) required to review a single paper. The graph demonstrates that as this cost rises, the system undergoes a two-stage degradation process, transitioning from perfect integrity to complete collapse.

The first transition occurs at the "Efficiency Limit" ($C_J \approx 0.23$). Below this value, the journal is efficient enough to process the entire potential volume of submissions (N) without compromising its screening rigor. Consequently, the threat of detection remains credible, and the system maintains the Honesty Regime ($Q = 1.0$). However, once C_J exceeds this limit, the journal enters the "Bistability Regime." Here, the cost of reviewing becomes too high to screen everyone, forcing the journal to lower its rationing probability ρ. This creates a "cracked door": the reduced probability of being reviewed (and thus detected) emboldens a fraction of opportunists to cheat ($0 < x^* < 1$). The curve in this yellow region visually represents this gradual erosion: as inefficiency (C_J) rises, the equilibrium fraction of bad papers increases, causing the average quality Q to decline smoothly.

The final blow comes at the "Collapse Point" ($C_J \approx 1.16$). Beyond this threshold, the review process becomes so expensive that the journal's effective capacity drops below the volume of even just the bad papers. The "Always Submit" strategy becomes the dominant equilibrium ($x^* > 1$), and the system falls into the Corruption Regime. In this red zone, the quality line flattens at the "Diluted Limit" ($Q \approx 0.24$). This flatness indicates that further inefficiency does not worsen the quality because the system is already maximally saturated with spam. The policy implication is stark: simply "funding" a journal is insufficient if the per-paper processing cost remains high. To restore integrity, institutions must invest in technologies or workflows that drastically reduce C_J, pushing the system back across the threshold into the efficient, honest regime.

4.10 Summary of Findings: The Dual Constraints of Capacity and Incentive

Our comprehensive sensitivity analysis reveals that the stability of the academic peer-review ecosystem is not governed by a single factor, but rather by the delicate interplay between two fundamental forces: physical capacity constraints and economic incentive structures. On the physical side, the journal's ability to maintain order is strictly limited by its processing throughput, determined by capacity K, review cost C_J, and friction ϵ. A counter-intuitive discovery in this domain is the "Productivity Paradox," where an increase in the production of high-quality research (α) ironically destabilizes the system by consuming scarce review resources. Similarly, the identification of a "Saturation Limit" demonstrates that structural inefficiencies (high C_J) cannot be overcome simply by eliminating cheaters; if the review process is too resource-intensive, even a fully honest population can overwhelm the system. These findings suggest that the sustainability of a journal depends as much on its operational efficiency and total submission volume (N) as it does on the ethical conduct of its authors. On the economic side, the parameters B, C, D, and λ define the "price" of stability. They determine the required screening intensity σ^* needed to deter opportunistic behavior. We observed that while high penalties (D) and submission costs (C) serve as effective filters that relax the burden on the screening system, high rewards (B) and false positive rates (λ) act as destabilizing amplifiers. Crucially, as the incentive to cheat increases, the system demands a disproportionately higher level of honesty from the community to reduce congestion and allow for stricter screening. When these economic incentives become too strong, they push the required honest fraction beyond physical possibilities ($x^* > 1$), leading to inevitable collapse regardless of capacity. Ultimately, the prevalence of the Bistability regime across wide parameter ranges underscores the critical role of path dependence. For a journal operating within standard constraints, the long-term outcome is rarely deterministic based on parameters alone but is heavily contingent on the initial state of the community. This implies that policy interventions must be designed not only to adjust parameters (e.g., increasing D or K) but also to actively steer the community's trust level (x) across the tipping point, as a system trapped in the basin of attraction for corruption cannot recover solely through marginal improvements in resources or penalties.

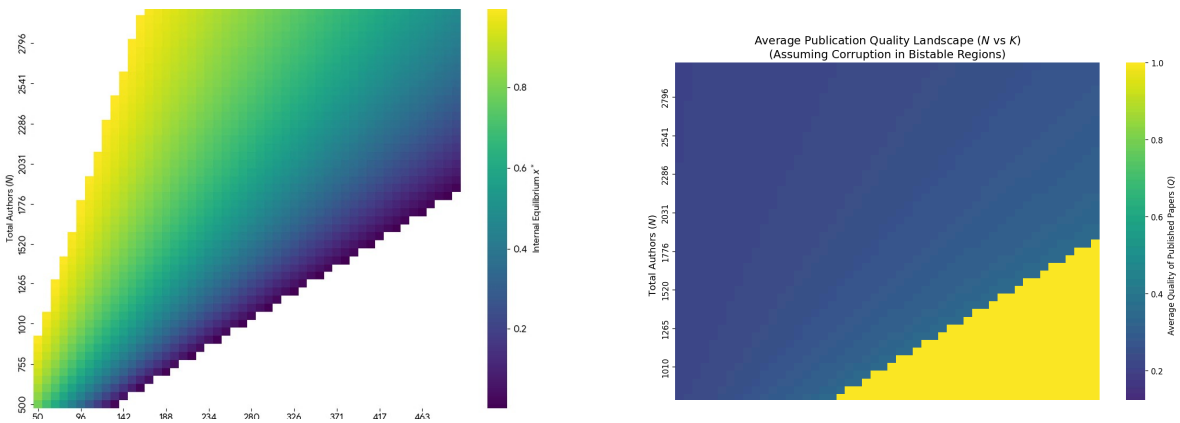
5 Joint Impact of Population and Capacity on Publication Quality

In this section, we extend our analysis from the internal stability equilibrium (x^*) to the macroscopic observable of the academic ecosystem: the *Average Quality of Published Papers* (Q). By varying the

total author population (N) and journal capacity (K) simultaneously, we construct a phase diagram that reveals the structural limits of the peer-review system. To map this quality landscape, we first determine the final evolutionary state of the population, denoted as x_{final}, based on the previously derived tipping point analysis. The system's behavior can be categorized into three distinct regimes based on the stability of the equilibrium. In the Global Honesty Regime ($x^* < 0$), the system possesses a single stable attractor at $x = 1$, implying that the final state is one of full honesty ($x_{final} = 1$). Conversely, in the Global Corruption Regime ($x^* > 1$), the single stable attractor lies at $x = 0$, leading to a final state of full corruption ($x_{final} = 0$). The intermediate Bistability Regime ($0 \leq x^* \leq 1$) presents a more complex scenario with two stable attractors. To evaluate the structural risk posed by congestion in this region, we adopt a "Worst-Case Scenario" approach, assuming that stochastic perturbations or insufficient initial trust will inevitably drive the system toward the corrupt attractor. Consequently, for the purpose of risk mapping, we assign $x_{final} = 0$ throughout the bistable region. Based on the determined final state x_{final}, we calculate the Average Quality (Q) of the accepted papers, defined as the ratio of valid scientific contributions to the total volume of accepted literature:

$$Q(x_{final}) = \frac{V_{good}^{acc}}{V_{good}^{acc} + V_{bad}^{acc}} \tag{64}$$

The volume of accepted papers depends critically on the submission behavior and the screening intensity σ. We assume that honest authors produce good papers at a rate α. Since the review process targets logical fallacies and misconduct rather than legitimate research, good papers are assumed to pass screening with probability 1, yielding a volume of $V_{good}^{acc} = N\alpha$. In contrast, opportunistic authors submit bad papers at a rate $(1-\alpha)$. These papers are accepted only if they successfully evade the screening process. The probability of evasion is given by $(1-\sigma)$, where $\sigma = \min\left(1, \frac{K}{C_J V_{total} + \epsilon}\right)$ is the realized screening probability constrained by capacity K and the total submission volume. Thus, the volume of accepted bad papers is expressed as $V_{bad}^{acc} = N(1-\alpha)(1-x_{final})(1-\sigma)$. We present the comparative analysis of the stability landscape and the resulting quality landscape in Figure 23. Figure 23(a) displays the internal equilibrium x^*, identifying the boundaries of stability, while Figure 23(b) visualizes the resulting average publication quality. A striking feature of the quality landscape is the sharp, non-smooth discontinuity where the average publication quality plunges abruptly from a perfect score of 1.0 to a low plateau of approximately 0.4. This "cliff-like" behavior represents a Phase Transition of the academic ecosystem, occurring when a macroscopic property of the system changes discontinuously as a control parameter crosses a critical threshold.



(a) Stability Landscape (x^*)

(b) Quality Landscape (Q)

Figure 23: Joint analysis of system stability and publication quality under varying Population (N) and Capacity (K). (a) The stability landscape shows the transition from Global Honesty (Green/Blue) to Global Corruption (Red) via a Bistable region. (b) The quality landscape reveals a sharp **Phase Transition**. The quality remains at a perfect 1.0 (Green) until the critical threshold is crossed, causing an instantaneous drop to a low plateau of ≈ 0.4 (Red), signifying the catastrophic collapse of the peer-review filter due to congestion.

The mechanism driving this phase transition can be understood by examining the system's response to congestion. In the "Safe Zone," where capacity K is sufficient relative to population N, the system's

attractor is $x_{final} = 1$. The submission pool consists exclusively of good papers, maintaining a perfect quality score of $Q = 1.0$. However, the moment the population N exceeds the critical carrying capacity defined by the contour $x^* = 0$, the system undergoes a catastrophic collapse, flipping to the corrupt attractor $x_{final} = 0$. This microscopic jump in strategy triggers a macroscopic explosion in submission volume from $N\alpha$ to N. In this corrupt state, the massive volume overwhelms the journal capacity, significantly diluting the screening probability σ. Consequently, the quality does not decay gradually to zero but settles at a "Diluted Limit" determined by the ratio of signal to noise. For instance, with $\alpha = 0.2$ and a saturated screening capability of $\sigma \approx 0.6$, the quality stabilizes at $Q_{corrupt} \approx \frac{\alpha}{\alpha+(1-\alpha)(1-\sigma)} \approx 0.38$. This analysis highlights a profound structural fragility: the system provides no early warning signs of impending failure. A journal can appear to be in perfect health with high-quality output while sitting immediately adjacent to the tipping point, where a minor increase in submissions or a slight reduction in budget can trigger an instantaneous and irreversible collapse in reputation.

A Supplementary Integrated Analysis

In this appendix, we present unified visualizations that integrate the individual parameter sensitivity analyses discussed in the main text. By plotting related variables on dual axes or within normalized ranges, we illustrate the comparative magnitude and directional impact of competing forces within the academic ecosystem.

A.1 Supply vs. Demand: Population and Capacity

Figure 24 contrasts the opposing effects of the demand-side pressure (Total Population N) and the supply-side constraint (Journal Capacity K). The dual x-axis format highlights a critical dynamic: increasing the population drives the system linearly toward corruption (Blue line), whereas increasing capacity provides a stabilizing force that restores honesty (Red dashed line). The intersection of these trends defines the sustainability limit of the journal.

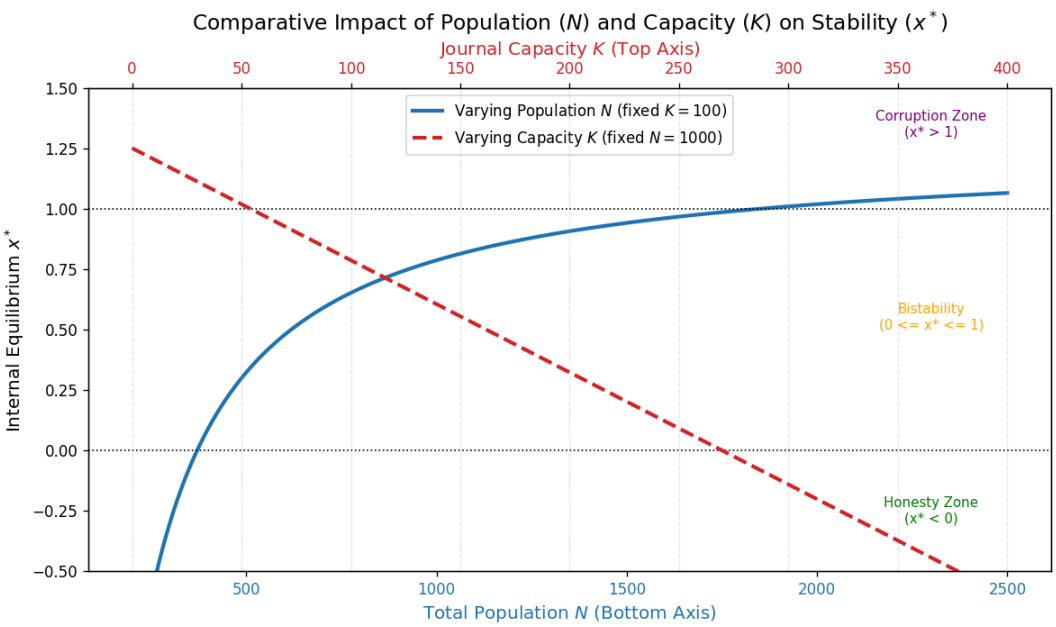


Figure 24: Comparative impact of Total Population (N, Bottom Axis) and Journal Capacity (K, Top Axis) on the internal equilibrium x^*. The opposing slopes illustrate how population growth acts as a destabilizing force (x^* increases), while capacity expansion acts as a stabilizing force (x^* decreases).

A.2 Normalized Parameter Sensitivity: Productivity, Friction, and Risk

Figure 25 integrates the sensitivity analysis for three parameters defined on the unit interval $[0, 1]$: Author Productivity (α), Screening Friction (ϵ), and False Positive Rate (λ). This visualization isolates the distinct functional forms of instability: ϵ introduces linear degradation, α exhibits hyperbolic growth characteristic of the "Productivity Paradox," and λ demonstrates a non-linear phase transition from unconditional honesty to rapid corruption.

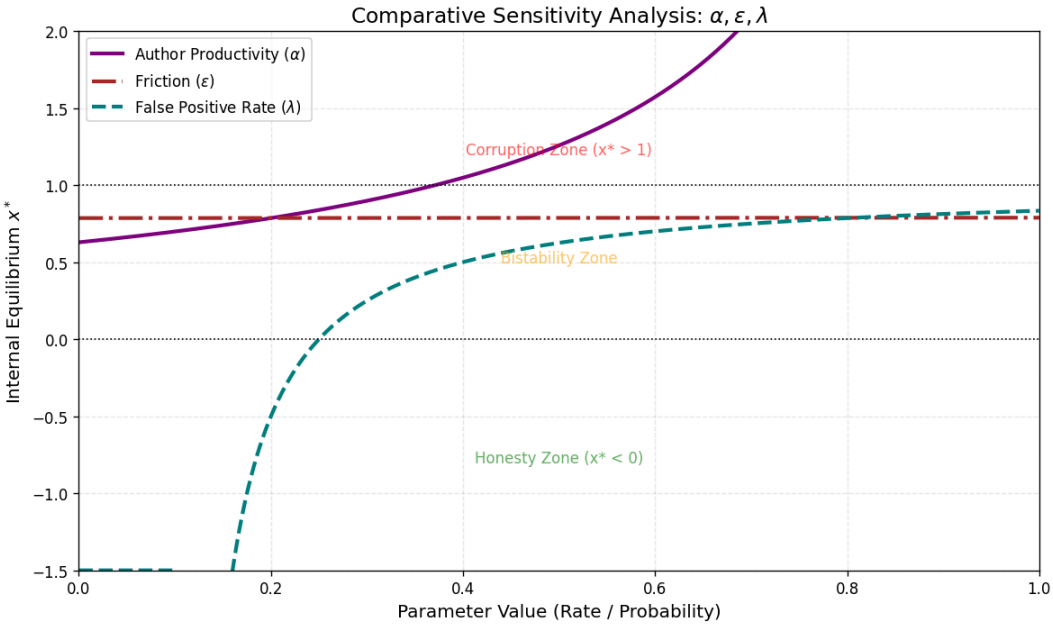


Figure 25: Unified sensitivity analysis for normalized parameters. The plot compares the linear impact of friction (ϵ, Brown), the hyperbolic impact of author productivity (α, Purple), and the threshold-based phase transition triggered by the false positive rate (λ, Teal).

A.3 Economic Forces: Incentive vs. Deterrent

Figure 26 visualizes the fundamental tug-of-war between the economic drivers of the system. The Benefit of Acceptance (B) serves as the primary incentive for opportunistic behavior, pulling the equilibrium x^* upward into the corruption zone. Conversely, the Penalty for Rejection (D) acts as the deterrent, suppressing x^* back toward the honesty zone. The graph demonstrates that maintaining stability requires the penalty D to scale proportionally with the benefit B.

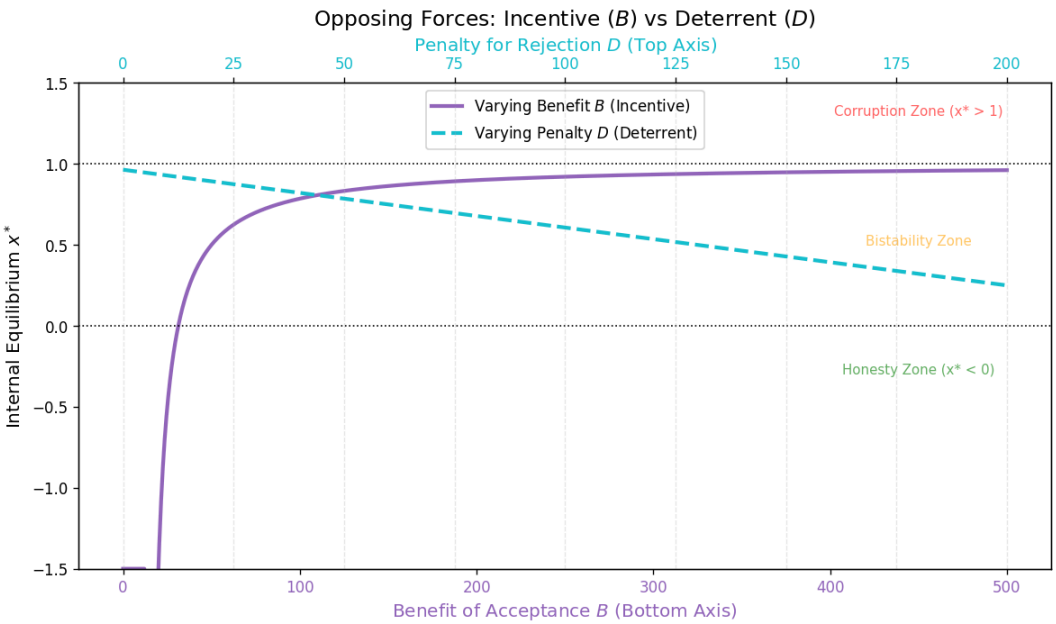


Figure 26: Comparative analysis of Economic Forces using a dual x-axis. The Purple line (Bottom Axis) shows how increasing the Benefit (B) destabilizes the system, while the Cyan dashed line (Top Axis) illustrates the stabilizing effect of increasing the Penalty (D).

A Model Parameters

The following table summarizes the key parameters used in the evolutionary game model, consistent with the empirical calibration and theoretical definitions discussed in the main text.

Symbol	Description	Role in Model
N	Total number of authors	Determines congestion intensity
K	Journal capacity	Sets the physical limit on publications
α	Good paper probability	Intrinsic productivity of authors
ϵ	False negative rate	Probability good papers are rejected
λ	False positive rate	Probability bad papers are accepted
r	Author reward	Incentive for publication
c	Submission cost	Barrier to entry for authors
B	Journal benefit	Utility from publishing good science
D	Journal penalty	Reputational cost of bad science
ρ	Rationing probability	Endogenous probability of acceptance

Table 4: Key notation and definitions for the Author-Journal Game.

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